Term-paper questions for Quantum Field Theory I

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June 29, 2026

I. PART A

Please answer all questions (i.e., 1., 2., and 3.) in this part.

1. Transformation properties of field bilinears in the (classical) Dirac theory, related to Section 2.5 of the lecture notes.

In this problem, you will consider again the Dirac theory and study the transformation properties of its physical observable under Lorentz transformations. Let

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu} \tag{0.1}$$

be a general Lorentz transformation, and $S(\Lambda)$ be the induced transformation for the Dirac spinors $\psi_a(x)$ (with $a = 1, \ldots, 4$):

$$\psi'_a(x') = S(\Lambda)_{ab}\psi_b(x). \tag{0.2}$$

Verify that the Dirac bilinears listed obey the following transformation laws:

(1) Scalar

$$\bar{\psi}'(x')\psi'(x') = \bar{\psi}(x)\psi(x). \tag{0.3}$$

By definition, the Dirac spinor transforms under a Lorentz transformation as $\psi'(x') = S(\Lambda)\psi(x)$. The adjoint spinor is $\bar{\psi} = \psi^\dagger\gamma^0$, which transforms as:

$$\bar{\psi}'(x') = \psi'^\dagger(x')\gamma^0 = \psi^\dagger(x)S(\Lambda)^\dagger\gamma^0.$$

Using the fundamental property of the spinor representation of the Lorentz group, $S(\Lambda)^\dagger\gamma^0 = \gamma^0 S(\Lambda)^{-1}$, we obtain:

$$\bar{\psi}'(x') = \psi^\dagger(x)\gamma^0 S(\Lambda)^{-1} = \bar{\psi}(x)S(\Lambda)^{-1}.$$

Therefore, the scalar bilinear transforms as:

$$\bar{\psi}'(x')\psi'(x') = \bar{\psi}(x)S(\Lambda)^{-1}S(\Lambda)\psi(x) = \bar{\psi}(x)\psi(x).$$

This proves that the scalar bilinear is a Lorentz invariant.

(2) Pseudoscalar

$$\bar{\psi}'(x')\gamma_5\psi'(x') = (\det\Lambda)\bar{\psi}(x)\gamma_5\psi(x). \tag{0.4}$$

Using $\bar{\psi}'(x') = \bar{\psi}(x)S(\Lambda)^{-1}$ and $\psi'(x') = S(\Lambda)\psi(x)$, the transformed pseudoscalar bilinear is:

$$\bar{\psi}'(x')\gamma_5\psi'(x') = \bar{\psi}(x)S(\Lambda)^{-1}\gamma_5 S(\Lambda)\psi(x).$$

The γ_5 matrix is defined by:

$$\gamma_5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = -\frac{i}{24}\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma,$$

where $\epsilon_{\mu\nu\rho\sigma}$ is the totally antisymmetric Levi-Civita tensor with $\epsilon_{0123} = 1$. Inserting this definition and using the relation $S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\alpha\gamma^\alpha$, we find:

$$\begin{aligned}
S(\Lambda)^{-1}\gamma_5 S(\Lambda) &= -\frac{i}{24}\epsilon_{\mu\nu\rho\sigma}S(\Lambda)^{-1}\gamma^\mu S(\Lambda)S(\Lambda)^{-1}\gamma^\nu S(\Lambda)S(\Lambda)^{-1}\gamma^\rho S(\Lambda)S(\Lambda)^{-1}\gamma^\sigma S(\Lambda) \\
&= -\frac{i}{24}\epsilon_{\mu\nu\rho\sigma}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta.
\end{aligned}$$

By the definition of the determinant of a 4×4 matrix, we have:

$$\epsilon_{\mu\nu\rho\sigma}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta = (\det\Lambda)\epsilon_{\alpha\beta\gamma\delta}.$$

Thus, we obtain:

$$S(\Lambda)^{-1}\gamma_5 S(\Lambda) = (\det\Lambda)\left(-\frac{i}{24}\epsilon_{\alpha\beta\gamma\delta}\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta\right) = (\det\Lambda)\gamma_5.$$

Substituting this back into the transformed bilinear yields:

$$\bar{\psi}'(x')\gamma_5\psi'(x') = (\det\Lambda)\bar{\psi}(x)\gamma_5\psi(x).$$

This proves that the pseudoscalar bilinear transforms as a Lorentz pseudoscalar.

(3) Vector

$$\bar{\psi}'(x')\gamma^\mu\psi'(x') = \Lambda^\mu{}_\nu\bar{\psi}(x)\gamma^\nu\psi(x). \tag{0.5}$$

Using the transformations of $\bar{\psi}'(x')$ and $\psi'(x')$, we write:

$$\bar{\psi}'(x')\gamma^\mu\psi'(x') = \bar{\psi}(x)S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\psi(x).$$

By the defining property of the spinor transformation $S(\Lambda)$, we have the relation:

$$S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu\gamma^\nu.$$

Applying this relation, we immediately obtain:

$$\bar{\psi}'(x')\gamma^\mu\psi'(x') = \Lambda^\mu{}_\nu\bar{\psi}(x)\gamma^\nu\psi(x).$$

This proves that the vector bilinear transforms as a Lorentz vector.

(4) Pseudovector

$$\bar{\psi}'(x')\gamma_5\gamma^\mu\psi'(x') = (\det\Lambda)\Lambda^\mu{}_\nu\bar{\psi}(x)\gamma_5\gamma^\nu\psi(x). \tag{0.6}$$

The transformed pseudovector bilinear is given by:

$$\bar{\psi}'(x')\gamma_5\gamma^\mu\psi'(x') = \bar{\psi}(x)S(\Lambda)^{-1}\gamma_5\gamma^\mu S(\Lambda)\psi(x).$$

We insert the identity $S(\Lambda)S(\Lambda)^{-1} = 1$ between γ_5 and γ^μ:

$$S(\Lambda)^{-1}\gamma_5\gamma^\mu S(\Lambda) = \left(S(\Lambda)^{-1}\gamma_5 S(\Lambda)\right)\left(S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\right).$$

Using the results from parts (2) and (3), namely $S(\Lambda)^{-1}\gamma_5 S(\Lambda) = (\det\Lambda)\gamma_5$ and $S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu\gamma^\nu$, we get:

$$S(\Lambda)^{-1}\gamma_5\gamma^\mu S(\Lambda) = (\det\Lambda)\gamma_5\Lambda^\mu{}_\nu\gamma^\nu = (\det\Lambda)\Lambda^\mu{}_\nu\gamma_5\gamma^\nu.$$

Thus, the bilinear transforms as:

$$\bar{\psi}'(x')\gamma_5\gamma^\mu\psi'(x') = (\det\Lambda)\Lambda^\mu{}_\nu\bar{\psi}(x)\gamma_5\gamma^\nu\psi(x).$$

(5) Tensor

$$\bar{\psi}'(x')\sigma^{\mu\nu}\psi'(x') = \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\bar{\psi}(x)\sigma^{\alpha\beta}\psi(x). \tag{0.7}$$

The tensor generator is defined as $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$. Under the transformation:

$$\bar{\psi}'(x')\sigma^{\mu\nu}\psi'(x') = \bar{\psi}(x)S(\Lambda)^{-1}\sigma^{\mu\nu}S(\Lambda)\psi(x).$$

We expand the commutator and insert $S(\Lambda)S(\Lambda)^{-1} = 1$ to obtain:

$$\begin{aligned}
S(\Lambda)^{-1}\sigma^{\mu\nu}S(\Lambda) &= \frac{i}{2}S(\Lambda)^{-1}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)S(\Lambda) \\
&= \frac{i}{2}\left[\left(S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\right)\left(S(\Lambda)^{-1}\gamma^\nu S(\Lambda)\right) - \left(S(\Lambda)^{-1}\gamma^\nu S(\Lambda)\right)\left(S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\right)\right].
\end{aligned}$$

Using $S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\alpha\gamma^\alpha$ and $S(\Lambda)^{-1}\gamma^\nu S(\Lambda) = \Lambda^\nu{}_\beta\gamma^\beta$, we get:

$$\begin{aligned}
S(\Lambda)^{-1}\sigma^{\mu\nu}S(\Lambda) &= \frac{i}{2}\left(\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\gamma^\alpha\gamma^\beta - \Lambda^\nu{}_\beta\Lambda^\mu{}_\alpha\gamma^\beta\gamma^\alpha\right) \\
&= \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\left(\frac{i}{2}[\gamma^\alpha, \gamma^\beta]\right) \\
&= \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\sigma^{\alpha\beta}.
\end{aligned}$$

Therefore, the tensor bilinear transforms as:

$$\bar{\psi}'(x')\sigma^{\mu\nu}\psi'(x') = \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\bar{\psi}(x)\sigma^{\alpha\beta}\psi(x).$$

Here, $\Lambda^\mu{}_\nu$ is a Lorentz transformation, and $\det\Lambda$ is its determinant.

2. Grassmann variables, related to Section 8.5 of the lecture notes.

(1) Let ξ and $\bar{\xi}$ be a pair of Grassmann variables. Let $g(\bar{\xi})$ be an analytic function of a single Grassmann variable $\bar{\xi}$, that is,

$$g(\bar{\xi}) = g_0 + g_1\bar{\xi} \tag{0.8}$$

and let $f(\xi)$ be another such function. Show that the inner product $\langle f|g\rangle$ defined by

$$\langle f|g\rangle := \int \mathrm{d}\bar{\xi}\,\mathrm{d}\xi\, e^{-\bar{\xi}\xi}\bar{f}(\xi)g(\bar{\xi}) \tag{0.9}$$

implies that

$$\langle f|g\rangle = \bar{f}_0 g_0 + \bar{f}_1 g_1, \tag{0.10}$$

where $\bar{f}_i$ stands for the complex conjugate of f_i.

Since $\xi^2 = 0$ and $\bar{\xi}^2 = 0$, any functions $f(\xi)$ and $g(\bar{\xi})$ can be expanded linearly as:

$$\bar{f}(\xi) = \bar{f}_0 + \bar{f}_1\xi, \quad g(\bar{\xi}) = g_0 + g_1\bar{\xi}.$$

The exponential term is expanded as:

$$e^{-\bar{\xi}\xi} = 1 - \bar{\xi}\xi = 1 + \xi\bar{\xi}.$$

Thus, the integrand is:

$$\begin{aligned} e^{-\bar{\xi}\xi}\bar{f}(\xi)g(\bar{\xi}) &= (1+\xi\bar{\xi})(\bar{f}_0 + \bar{f}_1\xi)(g_0 + g_1\bar{\xi}) \\ &= (1+\xi\bar{\xi})\left(\bar{f}_0 g_0 + \bar{f}_0 g_1\bar{\xi} + \bar{f}_1 g_0\xi + \bar{f}_1 g_1\xi\bar{\xi}\right). \end{aligned}$$

Using $\xi^2 = 0, \bar{\xi}^2 = 0$, the product becomes:

$$\bar{f}_0 g_0 + \bar{f}_0 g_1\bar{\xi} + \bar{f}_1 g_0\xi + \bar{f}_1 g_1\xi\bar{\xi} + \bar{f}_0 g_0\xi\bar{\xi},$$

where all other terms vanish. We can rewrite the quadratic terms as:

$$\bar{f}_1 g_1\xi\bar{\xi} + \bar{f}_0 g_0\xi\bar{\xi} = -(\bar{f}_0 g_0 + \bar{f}_1 g_1)\bar{\xi}\xi.$$

For Grassmann integration, the non-vanishing integrals are:

$$\int \mathrm{d}\bar{\xi}\mathrm{d}\xi\bar{\xi} = 0, \quad \int \mathrm{d}\bar{\xi}\mathrm{d}\xi\xi = 0, \quad \int \mathrm{d}\bar{\xi}\mathrm{d}\xi 1 = 0.$$

The only non-zero term under integration is the term proportional to $\bar{\xi}\xi$. Taking into account the anticommutation between the Grassmann variables (or between the variable and the integration measure), we have:

$$\int \mathrm{d}\bar{\xi}\mathrm{d}\xi\bar{\xi}\xi = \int \mathrm{d}\bar{\xi}\left(\int \mathrm{d}\xi(-\xi\bar{\xi})\right) = -\left(\int \mathrm{d}\bar{\xi}\bar{\xi}\right)\left(\int \mathrm{d}\xi\xi\right) = -1.$$

Therefore, the inner product is evaluated directly as:

$$\langle f|g\rangle = \int \mathrm{d}\bar{\xi}\mathrm{d}\xi \left[-(\bar{f}_0 g_0 + \bar{f}_1 g_1)\bar{\xi}\xi\right] = -(\bar{f}_0 g_0 + \bar{f}_1 g_1)(-1) = \bar{f}_0 g_0 + \bar{f}_1 g_1,$$

which proves the statement.

(2) Show that

$$(Af)(\bar{\xi}) = \int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\, A(\bar{\xi},\eta) f(\bar{\eta}) e^{-\bar{\eta}\eta} = g(\bar{\xi}) \tag{0.11}$$

is equivalent to

$$\begin{pmatrix} g_0 \\ g_1 \end{pmatrix} = \begin{pmatrix} A_{00} & A_{10} \\ A_{01} & A_{11} \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} \tag{0.12}$$

and that

$$(AB)(\bar{\xi},\xi) = \int \mathrm{d}\bar{\eta}\,\mathrm{d}\eta\, e^{-\bar{\eta}\eta} A(\bar{\xi},\eta) B(\bar{\eta},\xi) = C(\bar{\xi},\xi) \tag{0.13}$$

is equivalent to the standard definition of the product of two 2×2 matrices.

We expand the kernel $A(\bar{\xi},\eta)$ and the function $f(\bar{\eta})$ as:

$$A(\bar{\xi},\eta) = A_{00} + A_{01}\bar{\xi} + A_{10}\eta + A_{11}\bar{\xi}\eta, \quad f(\bar{\eta}) = f_0 + f_1\bar{\eta}.$$

The integral is:

$$(Af)(\bar{\xi}) = \int \mathrm{d}\bar{\eta}\mathrm{d}\eta \left(A_{00} + A_{01}\bar{\xi} + A_{10}\eta + A_{11}\bar{\xi}\eta\right)(f_0 + f_1\bar{\eta})(1 - \bar{\eta}\eta).$$

We expand the integrand and keep only terms containing both $\bar{\eta}$ and η (proportional to $\bar{\eta}\eta$), as all other terms integrate to zero:

- From the $-\bar{\eta}\eta$ term in $e^{-\bar{\eta}\eta}$: $(A_{00} + A_{01}\bar{\xi}) f_0 (-\bar{\eta}\eta) = -(A_{00} f_0 + A_{01} f_0 \bar{\xi})\bar{\eta}\eta$.
- From the 1 term in $e^{-\bar{\eta}\eta}$: $(A_{10}\eta + A_{11}\bar{\xi}\eta) f_1 \bar{\eta} = (A_{10} f_1 + A_{11} f_1 \bar{\xi})\eta\bar{\eta} = -(A_{10} f_1 + A_{11} f_1 \bar{\xi})\bar{\eta}\eta$.

Combining these terms, the non-vanishing part of the integrand is:

$$\text{Integrand} \supset -\left[(A_{00} f_0 + A_{10} f_1) + (A_{01} f_0 + A_{11} f_1)\bar{\xi}\right]\bar{\eta}\eta.$$

Integrating over $\mathrm{d}\bar{\eta}\mathrm{d}\eta$ using $\int \mathrm{d}\bar{\eta}\mathrm{d}\eta\bar{\eta}\eta = -1$ (as evaluated in part (1)), we obtain:

$$(Af)(\bar{\xi}) = g(\bar{\xi}) = g_0 + g_1\bar{\xi} = (A_{00} f_0 + A_{10} f_1) + (A_{01} f_0 + A_{11} f_1)\bar{\xi}.$$

Comparing the coefficients of 1 and $\bar{\xi}$ yields:

$$g_0 = A_{00} f_0 + A_{10} f_1, \quad g_1 = A_{01} f_0 + A_{11} f_1,$$

which is exactly equivalent to the matrix multiplication:

$$\begin{pmatrix} g_0 \\ g_1 \end{pmatrix} = \begin{pmatrix} A_{00} & A_{10} \\ A_{01} & A_{11} \end{pmatrix} \begin{pmatrix} f_0 \\ f_1 \end{pmatrix}.$$

Next, for the product of two operators $C = AB$:

$$C(\bar{\xi}, \xi) = \int \mathrm{d}\bar{\eta}\mathrm{d}\eta e^{-\bar{\eta}\eta} A(\bar{\xi}, \eta) B(\bar{\eta}, \xi).$$

Using the consistent expansions for the kernels:

$$A(\bar{\xi}, \eta) = A_{00} + A_{01}\bar{\xi} + A_{10}\eta + A_{11}\bar{\xi}\eta, \quad B(\bar{\eta}, \xi) = B_{00} + B_{01}\bar{\eta} + B_{10}\xi + B_{11}\bar{\eta}\xi.$$

Substituting these expansions into the integral and collecting the non-vanishing terms proportional to $\bar{\eta}\eta$, we find that the components of $C(\bar{\xi}, \xi) = C_{00} + C_{01}\bar{\xi} + C_{10}\xi + C_{11}\bar{\xi}\xi$ satisfy:

$$C_{ij} = \sum_k A_{ik} B_{kj},$$

which matches the standard definition of the product of two 2×2 matrices.

(3) Show that the operators $\hat{c}^\dagger$ and $\hat{c}$, defined by

$$\hat{c}^\dagger f(\bar{\xi}) := \bar{\xi} f(\bar{\xi}), \quad \hat{c} f(\bar{\xi}) := \frac{\mathrm{d}}{\mathrm{d}\bar{\xi}} f(\bar{\xi}), \tag{0.14}$$

satisfy canonical anticommutation relations, that is, $\hat{c}^\dagger \hat{c}^\dagger = \hat{c}\hat{c} = 0$ and $\{\hat{c}, \hat{c}^\dagger\} = 1$.

Let $f(\bar{\xi}) = f_0 + f_1\bar{\xi}$ be an arbitrary Grassmann function. 1. Anticommutation of $\hat{c}^\dagger$ with itself:

$$(\hat{c}^\dagger)^2 f(\bar{\xi}) = \hat{c}^\dagger(\bar{\xi} f(\bar{\xi})) = \bar{\xi}^2 f(\bar{\xi}).$$

Since Grassmann variables satisfy $\bar{\xi}^2 = 0$, we have $(\hat{c}^\dagger)^2 = 0$. 2. Anticommutation of $\hat{c}$ with itself:

$$\hat{c}^2 f(\bar{\xi}) = \frac{\mathrm{d}^2}{\mathrm{d}\bar{\xi}^2}(f_0 + f_1\bar{\xi}) = 0.$$

Thus $\hat{c}^2 = 0$. 3. Anticommutation $\{\hat{c}, \hat{c}^\dagger\} = \hat{c}\hat{c}^\dagger + \hat{c}^\dagger\hat{c}$: Applying the operators to $f(\bar{\xi})$:

$$\hat{c}^\dagger \hat{c} f(\bar{\xi}) = \hat{c}^\dagger \left(\frac{\mathrm{d}}{\mathrm{d}\bar{\xi}}(f_0 + f_1\bar{\xi}) \right) = \hat{c}^\dagger(f_1) = f_1\bar{\xi}.$$

$$\hat{c}\hat{c}^\dagger f(\bar{\xi}) = \hat{c}\left(\bar{\xi}(f_0 + f_1\bar{\xi})\right) = \hat{c}(f_0\bar{\xi}) = \frac{\mathrm{d}}{\mathrm{d}\bar{\xi}}(f_0\bar{\xi}) = f_0.$$

Adding the two results:

$$\{\hat{c}, \hat{c}^\dagger\} f(\bar{\xi}) = \left(\hat{c}\hat{c}^\dagger + \hat{c}^\dagger\hat{c}\right) f(\bar{\xi}) = f_0 + f_1\bar{\xi} = f(\bar{\xi}).$$

Since this holds for any $f(\bar{\xi})$, we have $\{\hat{c}, \hat{c}^\dagger\} = 1$.

(4) Show that, if $\{\bar{\xi}_j, \xi_j\}$ is a set of $2N$ Grassmann variables ($j = 1, \ldots, N$), then the Grassmann integral is

$$\int \left(\prod_{j=1}^{N} \mathrm{d}\bar{\xi}_j \, \mathrm{d}\xi_j \right) \exp\left\{ -\sum_{k,l=1}^{N} \bar{\xi}_k M_{kl} \xi_l \right\} = \det M. \tag{0.15}$$

We expand the exponential in the integrand:

$$\exp\left\{ -\sum_{k,l=1}^{N} \bar{\xi}_k M_{kl} \xi_l \right\} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\sum_{k,l=1}^{N} \bar{\xi}_k M_{kl} \xi_l \right)^n.$$

Since the integration measure contains $2N$ Grassmann differentials, only the term with $n = N$ survives integration. Terms with $n < N$ have fewer than $2N$ Grassmann variables and integrate to zero, while terms with $n > N$ vanish due to repeated Grassmann variables ($\xi_i^2 = 0, \bar{\xi}_i^2 = 0$). Thus, the integral reduces to:

$$I = \int \left(\prod_{j=1}^{N} \mathrm{d}\bar{\xi}_j \, \mathrm{d}\xi_j \right) \frac{(-1)^N}{N!} \sum_{\sigma,\tau \in S_N} M_{\sigma(1)\tau(1)} \ldots M_{\sigma(N)\tau(N)} \bar{\xi}_{\sigma(1)} \xi_{\tau(1)} \ldots \bar{\xi}_{\sigma(N)} \xi_{\tau(N)},$$

where σ and τ are permutations of $\{1, \ldots, N\}$.
We reorder the Grassmann product by collecting all $\bar{\xi}$ to the left and all ξ to the right. Interchanging the variables requires $N(N-1)/2$ anticommutations, yielding:

$$\bar{\xi}_{\sigma(1)} \xi_{\tau(1)} \ldots \bar{\xi}_{\sigma(N)} \xi_{\tau(N)} = (-1)^{N(N-1)/2} \bar{\xi}_{\sigma(1)} \ldots \bar{\xi}_{\sigma(N)} \xi_{\tau(1)} \ldots \xi_{\tau(N)}.$$

Next, sorting the variables to natural order $\{1, \ldots, N\}$ introduces the permutation signs:

$$\bar{\xi}_{\sigma(1)} \ldots \bar{\xi}_{\sigma(N)} \xi_{\tau(1)} \ldots \xi_{\tau(N)} = \mathrm{sgn}(\sigma)\mathrm{sgn}(\tau) \bar{\xi}_1 \ldots \bar{\xi}_N \xi_1 \ldots \xi_N.$$

To perform the sum over permutations, we rewrite the matrix product by reindexing using $\pi = \tau \circ \sigma^{-1}$ (so $\tau = \pi \circ \sigma$). Using $\mathrm{sgn}(\tau) = \mathrm{sgn}(\pi)\mathrm{sgn}(\sigma)$ and $(\mathrm{sgn}(\sigma))^2 = 1$, we get:

$$\sum_{\tau \in S_N} M_{\sigma(1)\tau(1)} \ldots M_{\sigma(N)\tau(N)} \mathrm{sgn}(\sigma)\mathrm{sgn}(\tau) = \sum_{\pi \in S_N} M_{1\pi(1)} \ldots M_{N\pi(N)} \mathrm{sgn}(\pi) = \det M.$$

Summing over all $\sigma \in S_N$ cancels the factor of $1/N!$ by producing $N!$ identical terms of $\det M$. Finally, performing the Grassmann integration, all sign factors $(-1)^N \times (-1)^{N(N-1)/2} \times (-1)^{N(N-1)/2} \times (-1)^{N^2} = (-1)^{2N + N(N-1)}$ combine to $+1$ since $2N$ and $N(N-1)$ are always even integers. Therefore, the integral evaluates to:

$$I = \det M.$$

3. Two-component complex scalar field, related to Chapters 3 and 4 of the lecture notes.

Consider the theory of a two-component complex scalar field $\phi_a(x)$ ($a = 1, 2$), which has the Lagrangian density

$$\mathcal{L} = (\partial_\mu \phi_a(x))^* (\partial^\mu \phi_a(x)) - V(\phi(x)), \tag{0.16}$$

where the potential reads

$$V(\phi(x)) = m_0^2 \phi_a^*(x)\phi_a(x), \tag{0.17}$$

and the repeated index a implies the summation over a. This theory is invariant under the classical global symmetry:

$$\begin{aligned} \phi_a(x) &\mapsto \phi'_a(x) = U_{ab}\phi_b(x), \\ \phi_a^*(x) &\mapsto \phi'^*_a(x) = U_{ab}^{-1}\phi_b^*(x), \\ \mathcal{L}(\phi') &= \mathcal{L}(\phi), \end{aligned} \tag{0.18}$$

where U is a 2×2 unitary matrix (i.e., $U^{-1} = U^\dagger$). This is the symmetry group $SU(2)$. Thus, ϕ transforms like the fundamental (spinor) representation of $SU(2)$. The matrices U can be expanded in the basis of 2×2 Pauli matrices $(\sigma_k)_{ab}$ and are parametrized by three Euler angles θ_k ($k = 1, 2, 3$):

$$U_{ab} = [\exp(i\theta_k\sigma_k)]_{ab} = \cos(|\boldsymbol{\theta}|)\delta_{ab} + i\frac{\boldsymbol{\theta}}{|\boldsymbol{\theta}|} \cdot \boldsymbol{\sigma}_{ab} \sin(|\boldsymbol{\theta}|). \tag{0.19}$$

(1) Use the classical canonical formalism to find (a) the canonical momentum π_a, conjugate to the field ϕ_a; (b) the Hamiltonian H; and (c) the total momentum P_j.

The Lagrangian density is:

$$\mathcal{L} = (\partial_\mu\phi_a)^*(\partial^\mu\phi_a) - m_0^2\phi_a^*\phi_a = \dot{\phi}_a^*\dot{\phi}_a - \nabla\phi_a^* \cdot \nabla\phi_a - m_0^2\phi_a^*\phi_a.$$

(a) The canonical momentum conjugate to the field $\phi_a(x)$ is defined by:

$$\pi_a(x) := \frac{\partial\mathcal{L}}{\partial\dot{\phi}_a} = \dot{\phi}_a^* = \partial^0\phi_a^*,$$

and the momentum conjugate to $\phi_a^*(x)$ is:

$$\pi_{\phi_a^*}(x) := \frac{\partial\mathcal{L}}{\partial\dot{\phi}_a^*} = \dot{\phi}_a = \pi_a^*.$$

(b) The Hamiltonian density is obtained via Legendre transformation:

$$\begin{aligned} \mathcal{H} &= \pi_a\dot{\phi}_a + \pi_{\phi_a^*}\dot{\phi}_a^* - \mathcal{L} \\ &= \pi_a\pi_a^* + \pi_a^*\pi_a - \left(\pi_a^*\pi_a - \nabla\phi_a^* \cdot \nabla\phi_a - m_0^2\phi_a^*\phi_a\right) \\ &= \pi_a^*\pi_a + \nabla\phi_a^* \cdot \nabla\phi_a + m_0^2\phi_a^*\phi_a. \end{aligned}$$

Thus, the total Hamiltonian is:

$$H = \int \mathrm{d}^3x \left(\pi_a^*\pi_a + \nabla\phi_a^* \cdot \nabla\phi_a + m_0^2\phi_a^*\phi_a\right).$$

(c) According to Noether's theorem for spacetime translation invariance $x'^\mu = x^\mu + a^\mu$, the conserved canonical energy-momentum tensor is defined by:

$$T^\mu_{\ \nu} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\partial_\nu\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a^*)}\partial_\nu\phi_a^* - \delta^\mu_{\ \nu}\mathcal{L}.$$

The total momentum P_j associated with spatial translations is given by:

$$P_j = -\int \mathrm{d}^3 x T^0_{\ j} = -\int \mathrm{d}^3 x \left(\pi_a^* \partial_j \phi_a + \pi_a \partial_j \phi_a^*\right).$$

(2) Derive the classical constants of motion associated with the global symmetry $SU(2)$. Relate these constants of motion to the generators of infinitesimal $SU(2)$ transformations. How many constants of motion do you find? Explain your results.

The Lagrangian is invariant under global $SU(2)$ transformations $\phi_a \to U_{ab}\phi_b$. An infinitesimal $SU(2)$ transformation parameterized by three parameters θ_k ($k = 1, 2, 3$) is:

$$U_{ab} \approx \delta_{ab} + i\theta_k(\sigma_k)_{ab}.$$

The infinitesimal variations of the fields are:

$$\delta\phi_a = i\theta_k(\sigma_k)_{ab}\phi_b, \quad \delta\phi_a^* = -i\theta_k(\sigma_k)_{ab}\phi_b^*.$$

According to Noether's theorem, the conserved currents associated with this continuous symmetry are:

$$J_k^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\delta_k\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a^*)}\delta_k\phi_a^*,$$

where $\delta_k\phi_a = i(\sigma_k)_{ab}\phi_b$ and $\delta_k\phi_a^* = -i(\sigma_k)_{ab}\phi_b^*$. The time component of the currents gives the conserved charges (constants of motion):

$$Q_k = \int \mathrm{d}^3 x J_k^0 = i\int \mathrm{d}^3 x \left(\pi_a(\sigma_k)_{ab}\phi_b - \pi_a^*(\sigma_k)_{ba}\phi_b^*\right).$$

Since the Lie algebra of $SU(2)$ is three-dimensional (generated by the three Pauli matrices $\sigma_1, \sigma_2, \sigma_3$), we find exactly 3 constants of motion.

1. Proof of conservation ($\frac{\mathrm{d}Q_k}{\mathrm{d}t} = 0$): Using the continuity equation $\partial_\mu J_k^\mu = \partial_0 J_k^0 + \nabla\cdot\mathbf{J}_k = 0$ and Gauss's theorem (assuming standard boundary conditions where fields vanish at infinity), we have:

$$\frac{\mathrm{d}Q_k}{\mathrm{d}t} = \int \mathrm{d}^3 x \partial_0 J_k^0 = -\int \mathrm{d}^3 x \nabla\cdot\mathbf{J}_k = -\int_\infty \mathbf{J}_k\cdot \mathrm{d}\mathbf{S} = 0.$$

This proves that Q_k are strict constants of motion.

2. Relation to generators via Poisson brackets: In classical field theory, a conserved charge Q_k generates the field transformations via canonical Poisson brackets. Using the fundamental Poisson brackets $\{\phi_a(\mathbf{x}), \pi_b(\mathbf{y})\}_{\mathrm{PB}} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$ and $\{\phi_a^*(\mathbf{x}), \pi_b^*(\mathbf{y})\}_{\mathrm{PB}} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$, we compute:

$$\{\phi_a(\mathbf{x}), \theta_k Q_k\}_{\mathrm{PB}} = i\theta_k \int \mathrm{d}^3 y \{\phi_a(\mathbf{x}), \pi_c(\mathbf{y})\}_{\mathrm{PB}}(\sigma_k)_{cd}\phi_d(\mathbf{y}) = i\theta_k(\sigma_k)_{ab}\phi_b(\mathbf{x}) = \delta\phi_a(\mathbf{x}),$$

$$\{\phi_a^*(\mathbf{x}), \theta_k Q_k\}_{\mathrm{PB}} = -i\theta_k \int \mathrm{d}^3 y \{\phi_a^*(\mathbf{x}), \pi_c^*(\mathbf{y})\}_{\mathrm{PB}}(\sigma_k)_{cd}\phi_d^*(\mathbf{y}) = -i\theta_k(\sigma_k)_{ab}\phi_b^*(\mathbf{x}) = \delta\phi_a^*(\mathbf{x}).$$

This explicitly demonstrates that the conserved charges Q_k are precisely the classical generators of infinitesimal $SU(2)$ transformations.

(3) Quantize this theory by imposing canonical commutation relations. Write an expression for the quantum mechanical Hamiltonian and total momentum operators in terms of the field and canonical momentum operators.

To quantize the classical theory obtained in part (1), we promote the classical field functions $\phi_a(\mathbf{x}), \phi_a^*(\mathbf{x})$ and their conjugate momenta $\pi_a(\mathbf{x}), \pi_a^*(\mathbf{x})$ to operators acting on the Hilbert space of states ($\phi_a \to \hat{\phi}_a, \phi_a^* \to \hat{\phi}_a^\dagger, \pi_a \to \hat{\pi}_a, \pi_a^* \to \hat{\pi}_a^\dagger$). Following Dirac's canonical quantization procedure, we replace the classical Poisson brackets with equal-time commutation relations:

$$[\phi_a(\mathbf{x}), \pi_b(\mathbf{y})] = i\delta_{ab}\delta^3(\mathbf{x}-\mathbf{y}), \quad [\phi_a^\dagger(\mathbf{x}), \pi_b^\dagger(\mathbf{y})] = i\delta_{ab}\delta^3(\mathbf{x}-\mathbf{y}),$$

with all other commutators vanishing. By substituting these operator definitions into the classical expressions for H and P_j derived in part (1), the quantum Hamiltonian and total momentum operators are expressed as:

$$H = \int \mathrm{d}^3x \left(\pi_a^\dagger \pi_a + \nabla\phi_a^\dagger \cdot \nabla\phi_a + m_0^2 \phi_a^\dagger \phi_a\right),$$

$$P_j = -\int \mathrm{d}^3x \left(\pi_a \partial_j \phi_a + \pi_a^\dagger \partial_j \phi_a^\dagger\right).$$

(4) Derive an expression for the quantum mechanical generators of global infinitesimal $SU(2)$ transformations in the Hilbert space of states of the system. Explain what relation, if any, they have with the conserved charges of the classical theory.

By applying canonical quantization to the classical $SU(2)$ Noether charges Q_k^{cl} derived in part (2), we promote the classical fields and conjugate momenta to quantum operators. This yields the quantum mechanical $SU(2)$ generators:

$$Q_k = i\int \mathrm{d}^3x \left(\pi_a(\sigma_k)_{ab}\phi_b - \pi_a^\dagger(\sigma_k)_{ba}\phi_b^\dagger\right).$$

Using the canonical commutation relations, the commutators of Q_k with the field operators evaluate to:

$$[Q_k, \phi_a(\mathbf{x})] = (\sigma_k)_{ab}\phi_b(\mathbf{x}), \quad [Q_k, \phi_a^\dagger(\mathbf{x})] = -\phi_b^\dagger(\mathbf{x})(\sigma_k)_{ba}.$$

Relation to the classical conserved charges derived in part (2):

1. **Direct Operator Equivalent via Canonical Quantization:** The quantum generators Q_k are the exact operator counterparts of the classical Noether charges Q_k^{cl} obtained in part (2). Under canonical quantization, the classical Poisson bracket generation rule $\{\phi_a, \theta_k Q_k^{\mathrm{cl}}\}_{\mathrm{PB}} = \delta\phi_a$ is directly mapped to the operator commutator rule $[Q_k, \phi_a] = (\sigma_k)_{ab}\phi_b$.

2. **Generators of Unitary Transformations in Hilbert Space:** While the classical charges Q_k^{cl} generate infinitesimal transformations on classical phase space, the Hermitian quantum

operators $Q_k = Q_k^\dagger$ generate unitary $SU(2)$ rotations $U(\boldsymbol{\theta}) = \exp(i\boldsymbol{\theta}\cdot\mathbf{Q})$ acting on state vectors in the Hilbert space.

3. **Preservation of Lie Algebra:** The quantum operators satisfy the exact $SU(2)$ Lie algebra $[Q_i, Q_j] = i\epsilon_{ijl}Q_l$, inheriting the Lie algebra structure of the Pauli matrices σ_k from the classical theory.

(5) Derive the quantum mechanical equations of motion of the Heisenberg representation operators.

In the Heisenberg representation, the equations of motion for any operator $\mathcal{O}(t)$ are given by $\dot{\mathcal{O}}(t) = i[H, \mathcal{O}(t)]$.
1. For the field operator $\phi_a(\mathbf{x}, t)$:

$$\dot{\phi}_a(\mathbf{x}, t) = i[H, \phi_a(\mathbf{x}, t)] = i\int \mathrm{d}^3y \left[\pi_b^\dagger(\mathbf{y}, t)\pi_b(\mathbf{y}, t), \phi_a(\mathbf{x}, t)\right] = \pi_a^\dagger(\mathbf{x}, t).$$

2. For the conjugate momentum operator $\pi_a(\mathbf{x}, t)$: Evaluating $[H, \pi_a(\mathbf{x}, t)]$, the spatial gradient term $\int \mathrm{d}^3y\, \nabla\phi_b^\dagger(\mathbf{y})\cdot\nabla\phi_b(\mathbf{y})$ involves an integration by parts to move the spatial derivatives onto the delta function $[\phi_b(\mathbf{y}), \pi_a(\mathbf{x})] = i\delta_{ab}\delta^3(\mathbf{y}-\mathbf{x})$, yielding $-i\nabla^2\phi_a^\dagger(\mathbf{x}, t)$. Combining this with the mass term gives:

$$\dot{\pi}_a(\mathbf{x}, t) = i[H, \pi_a(\mathbf{x}, t)] = \nabla^2\phi_a^\dagger(\mathbf{x}, t) - m_0^2\phi_a^\dagger(\mathbf{x}, t).$$

Taking the Hermitian conjugate and substituting $\dot{\phi}_a^\dagger = \pi_a$, the second time derivative of ϕ_a becomes:

$$\ddot{\phi}_a(\mathbf{x}, t) = \dot{\pi}_a^\dagger(\mathbf{x}, t) = \nabla^2\phi_a(\mathbf{x}, t) - m_0^2\phi_a(\mathbf{x}, t).$$

Rearranging terms, we obtain the Klein-Gordon equation for the operator:

$$(\partial^2 + m_0^2)\phi_a(x) = 0.$$

(6) Find an expansion of the field and canonical momentum operators in terms of a suitable set of creation and annihilation operators. How many species of creation and annihilation operators do you need? Justify your results.

Since $\phi_a(x)$ is a two-component $(a = 1, 2)$ complex scalar field, it describes particles and antiparticles for each component. Therefore, we need 4 distinct species of creation and annihilation operators:

- $a_a(\mathbf{k})$ and $a_a^\dagger(\mathbf{k})$: annihilates/creates type-a particles.
- $b_a(\mathbf{k})$ and $b_a^\dagger(\mathbf{k})$: annihilates/creates type-a antiparticles.

The mode expansions for the field and canonical momentum operators are:

$$\phi_a(\mathbf{x},t) = \int \frac{\mathrm{d}^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(a_a(\mathbf{k})e^{-ik\cdot x} + b_a^\dagger(\mathbf{k})e^{ik\cdot x}\right),$$

$$\pi_a(\mathbf{x},t) = i\int \frac{\mathrm{d}^3k}{(2\pi)^3}\sqrt{\frac{\omega_k}{2}} \left(a_a^\dagger(\mathbf{k})e^{ik\cdot x} - b_a(\mathbf{k})e^{-ik\cdot x}\right),$$

where $\omega_k = \sqrt{\mathbf{k}^2 + m_0^2}$. The canonical commutation relations for the fields are equivalent to the standard commutation relations for the creation and annihilation operators:

$$[a_a(\mathbf{k}), a_b^\dagger(\mathbf{k}')] = (2\pi)^3\delta_{ab}\delta^3(\mathbf{k}-\mathbf{k}'), \quad [b_a(\mathbf{k}), b_b^\dagger(\mathbf{k}')] = (2\pi)^3\delta_{ab}\delta^3(\mathbf{k}-\mathbf{k}').$$

All other commutators vanish.

(7) Find an expression for the $SU(2)$ generators in terms of creation and annihilation operators.

We substitute the mode expansions of $\phi_a(\mathbf{x},t)$, $\pi_a(\mathbf{x},t)$, $\phi_b^\dagger(\mathbf{x},t)$, and $\pi_a^\dagger(\mathbf{x},t)$ derived in part (6) into the quantum expression for the generators Q_k:

$$Q_k = i\int \mathrm{d}^3x \left(\pi_a(\mathbf{x},t)(\sigma_k)_{ab}\phi_b(\mathbf{x},t) - \pi_a^\dagger(\mathbf{x},t)(\sigma_k)_{ba}\phi_b^\dagger(\mathbf{x},t)\right) \equiv T_1 + T_2.$$

Expanding $T_1 = i\int \mathrm{d}^3x\,\pi_a(\sigma_k)_{ab}\phi_b$ and integrating over $\mathbf{x}$ using $\int \mathrm{d}^3x\, e^{\pm i(\mathbf{k}\mp\mathbf{p})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{k}\mp\mathbf{p})$, we obtain:

$$T_1 = -\frac{1}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}(\sigma_k)_{ab}\left[a_a^\dagger(\mathbf{k})a_b(\mathbf{k}) + a_a^\dagger(\mathbf{k})b_b^\dagger(-\mathbf{k})e^{2i\omega_k t} - b_a(\mathbf{k})a_b(-\mathbf{k})e^{-2i\omega_k t} - b_a(\mathbf{k})b_b^\dagger(\mathbf{k})\right].$$

Similarly, for $T_2 = -i\int \mathrm{d}^3x\,\pi_a^\dagger(\sigma_k)_{ba}\phi_b^\dagger$:

$$T_2 = -\frac{1}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}(\sigma_k)_{ba}\left[a_a(\mathbf{k})a_b^\dagger(\mathbf{k}) + a_a(\mathbf{k})b_b(-\mathbf{k})e^{-2i\omega_k t} - b_a^\dagger(\mathbf{k})a_b^\dagger(-\mathbf{k})e^{2i\omega_k t} - b_a^\dagger(\mathbf{k})b_b(\mathbf{k})\right].$$

Adding T_1 and T_2, the time-dependent cross terms proportional to $e^{\pm 2i\omega_k t}$ cancel out completely for all $k = 1,2,3$ upon changing variables $\mathbf{k}\to-\mathbf{k}$ and using $[a^\dagger, b^\dagger] = 0$.
For the remaining time-independent terms, applying the canonical commutation relations $[a_a(\mathbf{k}), a_b^\dagger(\mathbf{k}')] = (2\pi)^3\delta_{ab}\delta^3(\mathbf{k}-\mathbf{k}')$ and $[b_a(\mathbf{k}), b_b^\dagger(\mathbf{k}')] = (2\pi)^3\delta_{ab}\delta^3(\mathbf{k}-\mathbf{k}')$, the zero-point constant terms vanish due to the traceless property of the Pauli matrices, $\mathrm{Tr}(\sigma_k) = \sum_a(\sigma_k)_{aa} = 0$. Collecting the operator terms yields:

$$Q_k = -\int \frac{\mathrm{d}^3k}{(2\pi)^3}\left(a_a^\dagger(\mathbf{k})(\sigma_k)_{ab}a_b(\mathbf{k}) - b_a^\dagger(\mathbf{k})(\sigma_k)_{ba}b_b(\mathbf{k})\right).$$

Note on the overall sign: The overall negative sign arises directly from the problem statement's convention for the infinitesimal $SU(2)$ transformation $\phi_a'(x) = U_{ab}\phi_b(x) \approx (\delta_{ab} + i\theta_k(\sigma_k)_{ab})\phi_b(x)$

in Eq. (504), where $+i$ is chosen instead of $-i$. Here we retain the sign faithfully to the problem formulation.

(8) Find the ground state of the system and its quantum numbers. Find the Hamiltonian, the total momentum, and the $SU(2)$ generators normal-ordered relative to the ground state.

The ground state (vacuum) $|0\rangle$ of the system is defined by:

$$a_a(\mathbf{k})|0\rangle = 0, \quad b_a(\mathbf{k})|0\rangle = 0 \quad (\text{for all } a = 1, 2 \text{ and } \mathbf{k}).$$

Its quantum numbers are zero for all conserved quantities: energy $E = 0$, momentum $\mathbf{P} = \mathbf{0}$, and isospin charges $Q_k = 0$. The normal-ordered operators are obtained by moving all creation operators to the left of all annihilation operators:

$$:H: = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \omega_k \left(a_a^\dagger(\mathbf{k}) a_a(\mathbf{k}) + b_a^\dagger(\mathbf{k}) b_a(\mathbf{k}) \right),$$

$$:\mathbf{P}: = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \mathbf{k} \left(a_a^\dagger(\mathbf{k}) a_a(\mathbf{k}) + b_a^\dagger(\mathbf{k}) b_a(\mathbf{k}) \right),$$

$$:Q_k: = -\int \frac{\mathrm{d}^3k}{(2\pi)^3} \left(a_a^\dagger(\mathbf{k}) (\sigma_k)_{ab} a_b(\mathbf{k}) - b_a^\dagger(\mathbf{k}) (\sigma_k)_{ba} b_b(\mathbf{k}) \right).$$

(9) Find the spectrum of single particle states. Derive an expression for their energies, and assign quantum numbers to these states. Do you find any degeneracies? What is the degree of this degeneracy and why?

The single-particle and single-antiparticle states are created by acting on the vacuum state $|0\rangle$ with a single creation operator:

$$|p, a\rangle = a_a^\dagger(\mathbf{p})|0\rangle, \quad |\bar{p}, a\rangle = b_a^\dagger(\mathbf{p})|0\rangle \quad (a = 1, 2).$$

We determine the quantum numbers of these states by evaluating the action of the conserved operators $:H:$, $:\mathbf{P}:$, and $:Q_k:$ via their canonical commutation relations with creation operators.
1. Energy Eigenvalues (E): Using $[a_c(\mathbf{k}), a_a^\dagger(\mathbf{p})] = (2\pi)^3 \delta_{ca} \delta^3(\mathbf{k} - \mathbf{p})$ and $a_c(\mathbf{k})|0\rangle = 0$, the commutator with $:H:$ is:

$$[:H:, a_a^\dagger(\mathbf{p})] = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \omega_k a_c^\dagger(\mathbf{k}) [a_c(\mathbf{k}), a_a^\dagger(\mathbf{p})] = \omega_p a_a^\dagger(\mathbf{p}),$$

$$[:H:, b_a^\dagger(\mathbf{p})] = \int \frac{\mathrm{d}^3k}{(2\pi)^3} \omega_k b_c^\dagger(\mathbf{k}) [b_c(\mathbf{k}), b_a^\dagger(\mathbf{p})] = \omega_p b_a^\dagger(\mathbf{p}).$$

Acting on the vacuum yields:

$$:H: |p, a\rangle = \omega_p |p, a\rangle, \quad :H: |\bar{p}, a\rangle = \omega_p |\bar{p}, a\rangle.$$

Thus, both single-particle and single-antiparticle states have energy $E = \omega_p = \sqrt{\mathbf{p}^2 + m_0^2}$.
2. Momentum Quantum Numbers (p): Similarly, for the total momentum operator $:\mathbf{P}:$:

$$[:\mathbf{P}:, a_a^\dagger(\mathbf{p})] = \mathbf{p}a_a^\dagger(\mathbf{p}) \implies :\mathbf{P}: |p, a\rangle = \mathbf{p}|p, a\rangle,$$

$$[:\mathbf{P}:, b_a^\dagger(\mathbf{p})] = \mathbf{p}b_a^\dagger(\mathbf{p}) \implies :\mathbf{P}: |\bar{p}, a\rangle = \mathbf{p}|\bar{p}, a\rangle.$$

3. $SU(2)$ Generators / Isospin Quantum Numbers (Q_k): For the $SU(2)$ generators $:Q_k := -\int \frac{\mathrm{d}^3k}{(2\pi)^3} \Big(a_c^\dagger(\mathbf{k})(\sigma_k)_{cd}a_d(\mathbf{k}) - b_c^\dagger(\mathbf{k})(\sigma_k)_{dc}b_d(\mathbf{k})\Big):$

$$[:Q_k:, a_a^\dagger(\mathbf{p})] = -(\sigma_k)_{ca}a_c^\dagger(\mathbf{p}) \implies :Q_k: |p, a\rangle = -(\sigma_k)_{ca}|p, c\rangle,$$

$$[:Q_k:, b_a^\dagger(\mathbf{p})] = +(\sigma_k)_{ac}b_c^\dagger(\mathbf{p}) \implies :Q_k: |\bar{p}, a\rangle = +(\sigma_k)_{ac}|\bar{p}, c\rangle.$$

For the diagonal generator $k = 3$ ($\sigma_3 = \mathrm{diag}(1, -1)$):

- Particle states $|p, 1\rangle$ and $|p, 2\rangle$ have $Q_3 = -1$ and $Q_3 = +1$ respectively (forming an $SU(2)$ doublet with isospin $I = 1/2$, $I_3 = \mp 1/2$).
- Antiparticle states $|\bar{p}, 1\rangle$ and $|\bar{p}, 2\rangle$ have $Q_3 = +1$ and $Q_3 = -1$ respectively (forming the conjugate $SU(2)$ doublet).

4. $U(1)$ Charge (Q): Under the overall $U(1)$ charge $Q_{U(1)} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}(a_c^\dagger a_c - b_c^\dagger b_c)$, particle states have charge $Q = +1$ and antiparticle states have charge $Q = -1$.
5. Degeneracy: For any given momentum $\mathbf{p}$, there are 4 degenerate states: $|p, 1\rangle, |p, 2\rangle, |\bar{p}, 1\rangle, |\bar{p}, 2\rangle$, all sharing the exact same energy $E = \omega_p$. The 4-fold degeneracy arises from:

1. A factor of 2 from the internal $SU(2)$ isospin symmetry (components $a = 1, 2$).
2. A factor of 2 from particle-antiparticle charge conjugation symmetry ($a_a^\dagger$ versus $b_a^\dagger$).

Degree of degeneracy $= 2 \times 2 = 4$.

II. PART B

Please choose one question (i.e., 1. or 2.) in this part.

1. Feynman diagrams for a complex scalar field, related to Chapter 11 of the lecture notes.

In this problem, you will study the properties of a theory of a self-interacting complex scalar field $\phi(x)$, in perturbation theory. The Lagrangian density for this theory in four-dimensional Minkowski spacetime is

$$\mathcal{L} = \partial_\mu\phi^*(x)\partial^\mu\phi(x) - m_0^2\phi^*(x)\phi(x) - \lambda(\phi^*(x)\phi(x))^2 + J^*(x)\phi(x) + J(x)\phi^*(x) \qquad (0.20)$$

Assume that $m_0^2 > 0$ and that the coupling constant $\lambda > 0$. Here, the $J(x)$ are a set of (complex) sources. You are asked to do calculations using both canonical (operator) and path integral methods.

(1) Use path integral methods to derive an explicit expression for the generating function for the vacuum expectation values of time-ordered products $Z_0[J,J^*]$ for the free field theory in Minkowski spacetime and $Z_0[J,J^*]$ in Euclidean spacetime (imaginary time). Assume that the spacetime has infinite extent and that the sources (and the fields) vanish at infinity. Write your answer as an expression involving the sources $J(x)$, and a suitable vacuum expectation value of a time-ordered product of two free fields.

1. Microscopic Connection between Canonical Formalism and Path Integral In canonical quantization, the generating functional $Z_0[J,J^*]$ for the vacuum expectation values of time-ordered products of free complex scalar fields is defined by introducing complex external classical sources $J(x)$ and $J^*(x)$:

$$Z_0[J,J^*] \equiv \langle 0|\mathrm{T}\exp\left\{i\int \mathrm{d}^4x\left[J^*(x)\hat{\phi}(x)+J(x)\hat{\phi}^\dagger(x)\right]\right\}|0\rangle.$$

By Dyson's formula, the time-ordered product operator is equivalent to the full time-evolution operator $\hat{U}_J(t_f,t_i)=\exp\left(-i\hat{H}[J](t_f-t_i)\right)$ in the presence of the source-dependent Hamiltonian $\hat{H}[J]=\hat{H}_0-\int \mathrm{d}^3x[J^*(\mathbf{x})\hat{\phi}(\mathbf{x})+J(\mathbf{x})\hat{\phi}^\dagger(\mathbf{x})]$. Inserting completeness relations of field eigenstates $\int \mathcal{D}\phi_i\,|\phi_i\rangle\langle\phi_i|=1$ and $\int \mathcal{D}\phi_f\,|\phi_f\rangle\langle\phi_f|=1$ at initial and final times t_i,t_f yields:

$$\langle 0|e^{-i\hat{H}[J](t_f-t_i)}|0\rangle=\int \mathcal{D}\phi_f\mathcal{D}\phi_i\,\langle 0|\phi_f\rangle\underbrace{\langle\phi_f|e^{-i\hat{H}[J](t_f-t_i)}|\phi_i\rangle}_{\text{transition amplitude }\langle\phi_f,t_f|\phi_i,t_i\rangle_J}\langle\phi_i|0\rangle.$$

Slicing the time interval $[t_i,t_f]$ into N steps of length $\Delta t=(t_f-t_i)/N$, inserting field and conjugate momentum completeness relations $\int \mathcal{D}\pi_k\,|\pi_k\rangle\langle\pi_k|=1$ at each slice, and integrating out the momentum field π_k via Gaussian integration converts the Hamiltonian density $\mathcal{H}(\phi,\pi)-(J^*\phi+J\phi^*)$ into the classical Lagrangian action $i\int \mathrm{d}^4x[\mathcal{L}_0+J^*\phi+J\phi^*]$.
To project onto the pure vacuum state $|0\rangle$, we analytical-continue time into the complex plane via the $i\epsilon$ prescription, $t\to t(1-i\epsilon)$ with $t_f-t_i\to\infty$. Under this prescription, time evolution acquires a damping factor $e^{-\epsilon\hat{H}T}$, so all excited states $E_n>0$ decay exponentially to zero as $T\to\infty$, leaving only the ground state $|0\rangle$. The boundary wavefunctions $\langle 0|\phi_f\rangle\langle\phi_i|0\rangle$ are absorbed into an overall normalization constant $\mathcal{N}^{-1}$, rigorously establishing the path integral formulation in Minkowski spacetime:

$$Z_0[J,J^*]=\frac{1}{\mathcal{N}}\int \mathcal{D}\phi\mathcal{D}\phi^*\exp\left\{i\int \mathrm{d}^4x\left[\partial_\mu\phi^*\partial^\mu\phi-m_0^2\phi^*\phi+J^*\phi+J\phi^*+i\epsilon\phi^*\phi\right]\right\},$$

where $\mathcal{N}$ is chosen such that $Z_0[0,0]=1$.

2. Explicit Derivation via Completing the Square (Minkowski Spacetime) Integrating the kinetic term by parts (dropping boundary terms justified by the $i\epsilon$ damping at infinity), we re-express

the action in terms of the Klein-Gordon operator $\hat{K}_x \equiv -(\partial^2 + m_0^2 - i\epsilon)$:

$$S_0[J, J^*] = \int \mathrm{d}^4x \left[-\phi^*(x)(\partial^2 + m_0^2 - i\epsilon)\phi(x) + J^*(x)\phi(x) + J(x)\phi^*(x)\right].$$

The inverse operator $\hat{K}_x^{-1}$ corresponds to the Green's function $D_F(x-y)$ (Feynman propagator), which satisfies:

$$(\partial^2 + m_0^2 - i\epsilon)D_F(x-y) = -\delta^4(x-y) \quad \iff \quad \hat{K}_x D_F(x-y) = \delta^4(x-y),$$

with spatial symmetry $D_F(x-y) = D_F(y-x)$. To complete the square, we perform a linear shift of the integration variables toward the stationary point of the action:

$$\phi(x) = \phi_0(x) - \int \mathrm{d}^4y\, D_F(x-y)J(y), \qquad \phi^*(x) = \phi_0^*(x) - \int \mathrm{d}^4z\, J^*(z)D_F(z-x).$$

(Note: Taking the Hermitian adjoint of $\phi = -\hat{K}^{-1}J$ transposes the operator kernel as $(D_F^\dagger)_{zx} = D_F(z-x)^*$, giving $D_F(z-x)$ in $\phi^*(x)$ to maintain proper inner product ordering.)
Substituting these shifted fields into the exponent and expanding:

$$\begin{aligned}
&-\phi^*(x)\hat{K}_x\phi(x) + J^*(x)\phi(x) + J(x)\phi^*(x) \\
&= -\left(\phi_0^*(x) - \int \mathrm{d}^4zJ^*(z)D_F(z-x)\right)\hat{K}_x\left(\phi_0(x) - \int \mathrm{d}^4yD_F(x-y)J(y)\right) \\
&\quad + J^*(x)\phi_0(x) - J^*(x)\int \mathrm{d}^4yD_F(x-y)J(y) + J(x)\phi_0^*(x) - J(x)\int \mathrm{d}^4zJ^*(z)D_F(z-x).
\end{aligned}$$

Applying $\hat{K}_xD_F(x-y) = \delta^4(x-y)$ evaluates the operator action on the Green's function integrals:

$$\hat{K}_x \int \mathrm{d}^4y\, D_F(x-y)J(y) = -J(x), \qquad \int \mathrm{d}^4z\, J^*(z)D_F(z-x)\hat{K}_x = -J^*(x).$$

Thus, the cross-terms linear in ϕ_0 and ϕ_0^* cancel out completely, leaving the quadratic source term with a minus sign:

$$-\phi^*(x)\hat{K}_x\phi(x) + J^*(x)\phi(x) + J(x)\phi^*(x) = -\phi_0^*(x)\hat{K}_x\phi_0(x) - J^*(x)\int \mathrm{d}^4y\, D_F(x-y)J(y).$$

Because the shift $\delta\phi(x) = -\int \mathrm{d}^4yD_F(x-y)J(y)$ depends only on the fixed external source $J(y)$ (and is completely independent of the path integral variable ϕ_0), the functional derivative is $\dfrac{\delta\phi(x)}{\delta\phi_0(y)} = \delta^4(x-y) = \mathbb{I}$, so the functional Jacobian determinant is $\det(\mathbb{I}) = 1$:

$$\mathcal{D}\phi\mathcal{D}\phi^* = \mathcal{D}\phi_0\mathcal{D}\phi_0^*.$$

Furthermore, since the source-dependent exponent $-i\int \mathrm{d}^4x\mathrm{d}^4y\, J^*(x)D_F(x-y)J(y)$ contains no $\phi_0(x)$, it acts as a constant relative to the path integral $\int \mathcal{D}\phi_0\mathcal{D}\phi_0^*$ and factors out completely:

$$Z_0[J, J^*] = \left(\frac{1}{\mathcal{N}}\int \mathcal{D}\phi_0\mathcal{D}\phi_0^*\, e^{iS_0[\phi_0]}\right) \times \exp\left\{-i\int \mathrm{d}^4x\mathrm{d}^4y\, J^*(x)D_F(x-y)J(y)\right\}.$$

Normalizing the free field path integral to $Z_0[0,0]=1$, and identifying the 2-point time-ordered product $iD_F(x-y)=\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle$ (which means $-iD_F(x-y)=-\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle$), we obtain the exact closed-form expression:

$$Z_0[J,J^*]=\exp\left\{-i\int \mathrm{d}^4x\mathrm{d}^4y\, J^*(x)D_F(x-y)J(y)\right\}=\exp\left\{-\int \mathrm{d}^4x\mathrm{d}^4y\, J^*(x)\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^\dagger(y)|0\rangle J(y)\right\}.$$

3. Derivation in Euclidean Spacetime Under Wick rotation to imaginary time $\tau=it$ ($x^0=-ix_E^4, \partial_0=i\partial_4^E$), the Minkowski action transforms as $iS_M\to -S_E$, where the Euclidean action is positive-definite:

$$S_E[J_E,J_E^*]=\int \mathrm{d}^4x_E\left[(\partial_E\phi_E)^*(\partial_E\phi_E)+m_0^2\phi_E^*\phi_E-J_E^*\phi_E-J_E\phi_E^*\right].$$

Integrating by parts, $S_E=\int \mathrm{d}^4x_E\left[\phi_E^*(-\partial_E^2+m_0^2)\phi_E-J_E^*\phi_E-J_E\phi_E^*\right]$. The Euclidean Green's function $G_0(x_E-y_E)$ satisfies:

$$(-\partial_E^2+m_0^2)G_0(x_E-y_E)=\delta^4(x_E-y_E).$$

Performing the Euclidean field shift $\phi_E(x_E)=\phi_{E,0}(x_E)+\int \mathrm{d}^4y_E G_0(x_E-y_E)J_E(y_E)$, the linear terms in $\phi_{E,0}$ again cancel identically, giving:

$$-S_E=-\int \mathrm{d}^4x_E\,\phi_{E,0}^*(-\partial_E^2+m_0^2)\phi_{E,0}+\int \mathrm{d}^4x_E\mathrm{d}^4y_E J_E^*(x_E)G_0(x_E-y_E)J_E(y_E).$$

Factoring out the normalized free Euclidean path integral $\frac{1}{\mathcal{N}_E}\int \mathcal{D}\phi_{E,0}\mathcal{D}\phi_{E,0}^*e^{-S_E[\phi_{E,0}]}=1$, we obtain:

$$\begin{aligned}Z_0[J,J^*]&=\exp\left\{\int \mathrm{d}^4x_E\mathrm{d}^4y_E J^*(x_E)G_0(x_E-y_E)J(y_E)\right\}\\&=\exp\left\{\int \mathrm{d}^4x_E\mathrm{d}^4y_E J^*(x_E)\langle\hat{\phi}(x_E)\hat{\phi}^\dagger(y_E)\rangle_0 J(y_E)\right\}.\end{aligned}$$

where $\langle\hat{\phi}(x_E)\hat{\phi}^\dagger(y_E)\rangle_0=G_0(x_E-y_E)$ is the 2-point Euclidean correlation function.

(2) Use the path integral method to calculate, at the level of the free field theory (i.e., $\lambda = 0$) the following time-ordered products of the field $\hat{\phi}(x)$:

(a) $\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}(x')|0\rangle$,

(b) $\langle 0|\mathrm{T}\hat{\phi}^*(x)\hat{\phi}(x')|0\rangle$,

(c) $\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^*(x')|0\rangle$,

(d) $\langle 0|\mathrm{T}\hat{\phi}^*(x)\hat{\phi}^*(x')|0\rangle$.

Determine which of these expectation values are different from zero, and explain why. Use this result to define a set of Feynman propagators for this system. You can give your answer in terms of the Feynman propagator for a real scalar field, $G_0(x, x')$, which was discussed in Chapter 11.

We compute these expectation values by taking functional derivatives of the free generating functional $Z_0[J, J^*]$ with respect to the sources:

$$\langle 0|\mathrm{T}\hat{\phi}_1(x_1)\ldots\hat{\phi}_n(x_n)|0\rangle = \frac{1}{Z_0}\frac{\delta^n Z_0}{i\delta J^*(x_1)\ldots i\delta J(x_n)}\bigg|_{J=J^*=0}.$$

(a) For $\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}(x')|0\rangle$, we take derivatives with respect to $J^*(x)$ and $J^*(x')$:

$$\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}(x')|0\rangle = \frac{\delta^2 Z_0}{i\delta J^*(x)i\delta J^*(x')}\bigg|_{J=J^*=0} = 0.$$

(d) Similarly, for $\langle 0|\mathrm{T}\hat{\phi}^*(x)\hat{\phi}^*(x')|0\rangle$, we take derivatives with respect to $J(x)$ and $J(x')$:

$$\langle 0|\mathrm{T}\hat{\phi}^*(x)\hat{\phi}^*(x')|0\rangle = \frac{\delta^2 Z_0}{i\delta J(x)i\delta J(x')}\bigg|_{J=J^*=0} = 0.$$

These expectation values vanish because the free Lagrangian has a global $U(1)$ symmetry $\phi \to e^{i\theta}\phi$, which protects the charge. Since ϕ has charge $+1$ and ϕ^* has charge -1, VEVs of operators with non-zero net charge must vanish. (b) & (c) For the mixed VEVs, we obtain:

$$\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^*(x')|0\rangle = \langle 0|\mathrm{T}\hat{\phi}^*(x')\hat{\phi}(x)|0\rangle = \frac{\delta^2 Z_0}{i\delta J^*(x)i\delta J(x')}\bigg|_{J=J^*=0} = iD_F(x - x').$$

For a complex scalar field, we can decompose it into two real scalar fields ϕ_1, ϕ_2 as $\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$. The real fields satisfy $\langle 0|\mathrm{T}\phi_i(x)\phi_j(x')|0\rangle = \delta_{ij}G_0(x, x')$, where $G_0(x, x')$ is the real scalar Feynman propagator. Thus:

$$\langle 0|\mathrm{T}\hat{\phi}(x)\hat{\phi}^*(x')|0\rangle = \frac{1}{2}\langle 0|\mathrm{T}(\phi_1(x) + i\phi_2(x))(\phi_1(x') - i\phi_2(x'))|0\rangle = G_0(x, x').$$

Hence, the non-zero mixed propagator is exactly equal to the Feynman propagator of a real scalar field, $G_0(x, x')$.

(3) Use path integral methods to derive a formula for the generating function $Z[J, J^*]$ of the full interacting theory in terms of $Z_0[J, J^*]$ and of the interaction part of the Lagrangian. Use this formula to derive the Feynman rules for this theory both in Minkowski spacetime and in Euclidean spacetime.

The full interacting path integral $Z[J, J^*]$ is defined as:

$$Z[J, J^*] = \frac{1}{\mathcal{N}} \int \mathcal{D}\phi\mathcal{D}\phi^* \exp\left\{ i \int \mathrm{d}^4x \left[\mathcal{L}_{\text{free}}(\phi, \phi^*) - \lambda(\phi^*\phi)^2 + J^*\phi + J\phi^* \right] \right\}.$$

To factor out the interaction part, we split the exponent into the interaction term and the free-field plus source term:

$$Z[J, J^*] = \frac{1}{\mathcal{N}} \int \mathcal{D}\phi\mathcal{D}\phi^* \exp\left\{ -i \int \mathrm{d}^4x \lambda(\phi^*(x)\phi(x))^2 \right\} \exp\left\{ iS_0[\phi, \phi^*] + i \int \mathrm{d}^4x(J^*\phi + J\phi^*) \right\}.$$

Under the path integral, functional derivatives of the source coupling term $E[J, J^*] \equiv \exp\left\{ i \int \mathrm{d}^4y(J^*\phi + J\phi^*) \right\}$ with respect to the sources yield the classical field components:

$$\frac{\delta}{i\delta J^*(x)} E[J, J^*] = \phi(x)E[J, J^*], \qquad \frac{\delta}{i\delta J(x)} E[J, J^*] = \phi^*(x)E[J, J^*].$$

Iterating this derivative rule for any functional $\mathcal{F}[\phi, \phi^*]$ allows us to replace field variables inside the path integral by functional differential operators:

$$\mathcal{F}[\phi, \phi^*] \exp\left\{ iS_0 + i \int (J^*\phi + J\phi^*) \right\} = \mathcal{F}\left[\frac{\delta}{i\delta J^*}, \frac{\delta}{i\delta J} \right] \exp\left\{ iS_0 + i \int (J^*\phi + J\phi^*) \right\}.$$

Specifically, for the interaction density $(\phi^*(x)\phi(x))^2$, the field operators are mapped to:

$$(\phi^*(x)\phi(x))^2 \longleftrightarrow \left(\frac{\delta}{i\delta J(x)} \frac{\delta}{i\delta J^*(x)} \right)^2.$$

Because the resulting differential operator $\exp\left\{ -i \int \mathrm{d}^4x\lambda \left(\frac{\delta}{i\delta J(x)} \frac{\delta}{i\delta J^*(x)} \right)^2 \right\}$ acts purely on the sources J, J^* and is independent of the integration variables ϕ, ϕ^*, it can be factored completely out of the functional integral $\int \mathcal{D}\phi\mathcal{D}\phi^*$:

$$\begin{aligned} Z[J, J^*] &= \exp\left\{ -i \int \mathrm{d}^4x\lambda \left(\frac{\delta}{i\delta J(x)} \frac{\delta}{i\delta J^*(x)} \right)^2 \right\} \underbrace{\left(\frac{1}{\mathcal{N}} \int \mathcal{D}\phi\mathcal{D}\phi^* \exp\left\{ iS_0 + i \int (J^*\phi + J\phi^*) \right\} \right)}_{=Z_0[J,J^*]} \\ &= \exp\left\{ -i \int \mathrm{d}^4x\lambda \left(\frac{\delta}{i\delta J(x)} \frac{\delta}{i\delta J^*(x)} \right)^2 \right\} Z_0[J, J^*]. \end{aligned}$$

In Euclidean spacetime (where $t = -i\tau$, $i\mathrm{d}^4x \to -\mathrm{d}^4x_E$, and $\frac{\delta}{i\delta J} \to \frac{\delta}{\delta J_E}$), the corresponding formula becomes:

$$Z_E[J, J^*] = \exp\left\{-\int \mathrm{d}^4x_E \lambda \left(\frac{\delta}{\delta J(x_E)}\frac{\delta}{\delta J^*(x_E)}\right)^2\right\} Z_{0,E}[J, J^*].$$

Note on normalization: In free field theory, $Z_0[J, J^*]$ is normalized such that $Z_0[0, 0] = 1$. In the interacting theory, however, setting $J = J^* = 0$ in the unnormalized functional gives $Z[0, 0] = Z_{\text{vacuum}} \neq 1$ (the sum of all disconnected vacuum bubble diagrams). Consequently, physical n-point correlation functions in the full theory must be computed using the normalized functional derivative $\frac{1}{Z[0,0]} \frac{\delta^n Z[J, J^*]}{i^n \delta J^* \ldots \delta J \ldots}\bigg|_{J=0}$, where the denominator $\frac{1}{Z[0,0]}$ serves to cancel all vacuum diagrams as proven in (4).
To derive the Feynman rules, we expand the exponential interaction operator in a Taylor series in powers of λ:

$$Z[J, J^*] = \sum_{V=0}^{\infty} \frac{1}{V!}\left[-i\lambda \int \mathrm{d}^4x \left(\frac{\delta}{i\delta J(x)}\right)^2 \left(\frac{\delta}{i\delta J^*(x)}\right)^2\right]^V Z_0[J, J^*].$$

Each order V corresponds to a diagram with V interaction vertices located at spacetime points $x_1, \ldots, x_V$. Applying the derivatives to the free field generating functional $Z_0[J, J^*] = \exp\left\{-i\int J^* D_F J\right\}$ generates the graphical elements of the perturbation theory:

- **Propagator (Internal Line)**: Differentiating $Z_0[J, J^*]$ with respect to a pair of sources $J^*(x)$ and $J(y)$ extracts the 2-point Green's function:

$$\frac{\delta^2 Z_0[J, J^*]}{i\delta J^*(x) i\delta J(y)}\bigg|_{J=J^*=0} = iD_F(x - y).$$

  In momentum space, this gives a line carrying momentum p representing $iD_F(p) = \frac{i}{p^2 - m_0^2 + i\epsilon}$ (Minkowski) or $G_0(p) = \frac{1}{p_E^2 + m_0^2}$ (Euclidean). Because $\frac{\delta}{i\delta J^*(x)}$ corresponds to $\phi(x)$ (charge $+1$) and $\frac{\delta}{i\delta J(y)}$ corresponds to $\phi^*(y)$ (charge -1), the propagator is drawn with a directed arrow pointing from y to x, representing the flow of $U(1)$ charge.

- **Interaction Vertex**: At each vertex x, the operator $-i\lambda\left(\frac{\delta}{i\delta J(x)}\right)^2\left(\frac{\delta}{i\delta J^*(x)}\right)^2$ contains 2 derivatives with respect to J (incoming lines) and 2 derivatives with respect to J^* (outgoing lines), forming a 4-point vertex. Permuting the 2 identical incoming lines ($2! = 2$) and the 2 identical outgoing lines ($2! = 2$) yields a symmetry/combinatorial factor of $2! \times 2! = 4$. Multiplying by $-i\lambda$ gives the Minkowski vertex factor $\mathbf{-i4}\lambda$ (or $\mathbf{-4}\lambda$ in Euclidean spacetime).

- **External Lines**: Differentiating $Z[J, J^*]$ with respect to external sources $J^*(z)$ (for a field $\phi(z)$) or $J(z)$ (for $\phi^*(z)$) connects the external spacetime points to the vertices via free propagators.

(4) Give a general argument to prove that the vacuum diagrams cancel out from the computation of vacuum expectation values of any number of field operators.

The generating functional of the interacting theory $Z[J, J^*]$ can be written as:

$$Z[J, J^*] = (\text{Sum of all Feynman diagrams with sources } J, J^*)\,.$$

Any diagram in this expansion can be decomposed into a set of connected diagrams that are attached to external sources, and a set of closed loop diagrams that are completely disconnected from any external sources (vacuum diagrams). Therefore, we can write:

$$Z[J, J^*] = Z_{\text{connected}}[J, J^*]Z_{\text{vacuum}},$$

where $Z_{\text{vacuum}} = Z[0, 0]$ is the sum of all vacuum diagrams. The vacuum expectation value of any product of operators is given by the normalized functional derivative:

$$\langle 0|\mathrm{T}\hat{\phi}(x_1)\ldots\hat{\phi}^*(y_1)\ldots|0\rangle = \frac{1}{Z[0,0]}\left(\frac{\delta}{i\delta J^*(x_1)}\cdots\frac{\delta}{i\delta J(y_1)}\cdots Z[J, J^*]\right)\Bigg|_{J=0}.$$

Substituting the factored form of $Z[J, J^*]$, we find:

$$\langle 0|\mathrm{T}\hat{\phi}(x_1)\ldots\hat{\phi}^*(y_1)\ldots|0\rangle = \left(\frac{\delta}{i\delta J^*(x_1)}\cdots\frac{\delta}{i\delta J(y_1)}\cdots Z_{\text{connected}}[J, J^*]\right)\Bigg|_{J=0}.$$

The multiplicative factor $Z[0, 0]$ in the numerator and denominator cancels out completely. This shows that vacuum diagrams do not contribute to physical correlation functions and can be ignored.

(5) Use the Feynman rules that you derived in (1) of this exercise to obtain the perturbation theory expansion in Euclidean spacetime for

(a) the vacuum diagrams,

(b) the two-point functions,

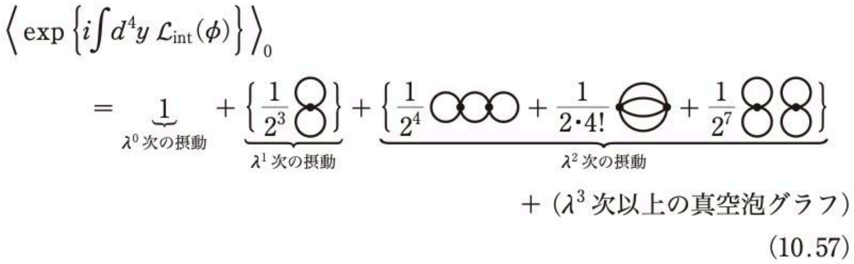
(c) the four-point functions,

in position space and up to and including all terms up to second order in the coupling constant λ. Assign a Feynman diagram to each term. Give an explicit formula for each term, including the multiplicity factors. Do not do the integrals! Verify the cancellation of the vacuum diagrams.

Note on Reference and Theoretical / Notational Differences The Feynman diagrams presented in this solution are adapted from Chapter 10, "Perturbation Calculations and Feynman Graphs", of *Quantum Field Theory (II) – Focusing on Feynman Graphs and Renormalization* by Masato Sakamoto (Quantum Mechanics Series, Shokabo, 2020).
Note that the referenced textbook considers a real scalar field (ϕ^4 theory), and thus the prop-

agators in the original figures are drawn without directional arrows. In contrast, the model studied in this problem (Part B, Question 1) is a complex scalar field ($\lambda(\phi^*\phi)^2$ theory), where $U(1)$ charge conservation requires directed arrows on propagators, with two incoming and two outgoing lines at each interaction vertex. This constraint on line orientations alters the number of valid Wick contractions and symmetry factors (combinatorial factors) compared to real scalar graphs of identical topology. However, to maintain clarity and avoid confusion, the expansion coefficients and notations in the formulas below are written in direct one-to-one correspondence with the figures referenced from Chapter 10 of Sakamoto's book. **(Note: The complete classification of directed Feynman graphs (oriented graphs) with arrow orientations and exact symmetry factors for the complex scalar field is also provided at the end of the answer to this subquestion (5).)**
Regarding the propagator notation, the Green's function $D_F(x-y)$ defined in Minkowski spacetime in parts (1)–(3) (satisfying $(\partial^2+m_0^2-i\epsilon)D_F(x-y) = -\delta^4(x-y)$) is related to the two-point time-ordered expectation value of free fields by $\langle 0|\mathrm{T}\phi(x)\phi^\dagger(y)|0\rangle = iD_F(x-y)$. Hence, the free propagator notation $\Delta_{F,0}(x-y) \equiv \langle 0|\mathrm{T}\phi(x)\phi(y)|0\rangle_0$ used in the referenced figures differs by a factor of the imaginary unit i, namely $\Delta_{F,0}(x-y) = iD_F(x-y)$. Furthermore, in Euclidean spacetime as requested in this question (5), Wick rotation ($t = -i\tau$) eliminates the factor of i, making it equivalent to the real Euclidean propagator $G_0(x_E - y_E)$ (with momentum space representation $\dfrac{1}{k_E^2+m_0^2}$) introduced in parts (1) and (2).

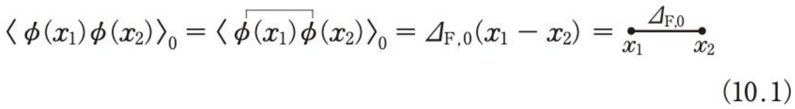
(a) Vacuum Diagrams



$$\langle \exp\left\{i\int \mathrm{d}^4y\,\mathcal{L}_{\rm int}(\phi)\right\}\rangle_0 = 1 + \frac{1}{2^3}(-i\lambda)\int \mathrm{d}^4y\,(\Delta_{F,0}(0))^2$$
$$+ (-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\left[\frac{1}{2^4}(\Delta_{F,0}(0))^4 + \frac{1}{2\cdot 4!}\Delta_{F,0}(0)(\Delta_{F,0}(y_1-y_2))^2\Delta_{F,0}(0)\right.$$
$$\left.+\frac{1}{2^7}(\Delta_{F,0}(y_1-y_2))^4\right] + \mathcal{O}(\lambda^3).$$

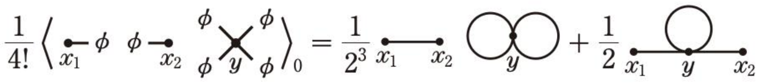
(b) Two-Point Functions

Order λ^0 ($k=0$)



$$G^{(2)}_{k=0}(x_1,x_2) = \Delta_{F,0}(x_1-x_2).$$

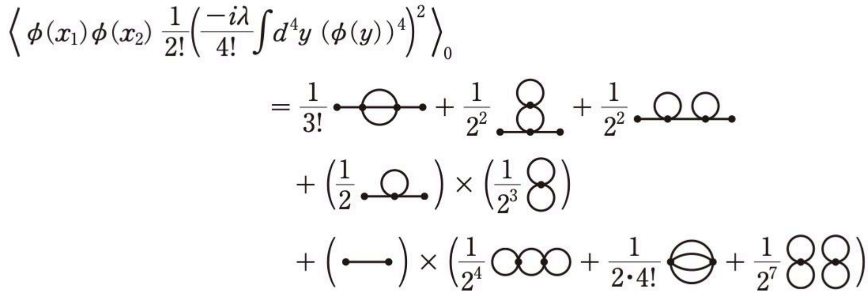
Order λ^1 ($k=1$)



(10.14)

$$G^{(2)}_{k=1}(x_1,x_2) = \frac{1}{2^3}(-i\lambda)\int \mathrm{d}^4y\,\Delta_{F,0}(x_1-x_2)(\Delta_{F,0}(0))^2$$
$$+\frac{1}{2}(-i\lambda)\int \mathrm{d}^4y\,\Delta_{F,0}(x_1-y)\Delta_{F,0}(0)\Delta_{F,0}(y-x_2).$$

Order λ^2 ($k=2$)



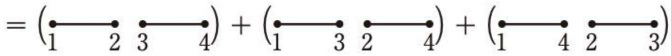
(10.28)

$$G^{(2)}_{k=2}(x_1,x_2) = \frac{1}{3!}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)(\Delta_{F,0}(y_1-y_2))^3\Delta_{F,0}(y_2-x_2)$$
$$+\frac{1}{2^2}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)\Delta_{F,0}(0)\Delta_{F,0}(y_1-y_2)\Delta_{F,0}(0)\Delta_{F,0}(y_2-x_2)$$
$$+\frac{1}{2^2}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)(\Delta_{F,0}(y_1-y_2))^2\Delta_{F,0}(y_2-x_2)\Delta_{F,0}(0)$$
$$+\left[\frac{1}{2}(-i\lambda)\int \mathrm{d}^4y_1\,\Delta_{F,0}(x_1-y_1)\Delta_{F,0}(0)\Delta_{F,0}(y_1-x_2)\right]\left[\frac{1}{2^3}(-i\lambda)\int \mathrm{d}^4y_2\,(\Delta_{F,0}(0))^2\right]$$
$$+\Delta_{F,0}(x_1-x_2)\times(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\left[\frac{1}{2^4}\Delta_{F,0}(0)(\Delta_{F,0}(y_1-y_2))^2\Delta_{F,0}(0)\right.$$
$$\left.+(\Delta_{F,0}(y_1-y_2))^4+\frac{1}{2^7}(\Delta_{F,0}(y_1-y_2))^4\right].$$

(c) Four-Point Functions

Order λ^0 ($k=0$)

$$\langle \phi_1\phi_2\phi_3\phi_4 \rangle_0 = \langle \phi_1\phi_2\phi_3\phi_4 \rangle_0 + \langle \phi_1\phi_2\phi_3\phi_4 \rangle_0 + \langle \phi_1\phi_2\phi_3\phi_4 \rangle_0$$
$$= \Delta_{12}\Delta_{34} + \Delta_{13}\Delta_{24} + \Delta_{14}\Delta_{23}$$

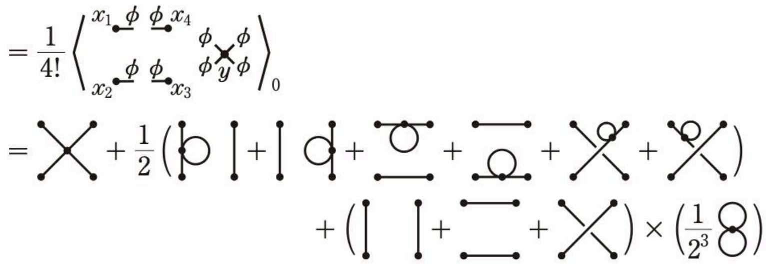


(10.2)

$$G^{(4)}_{k=0}(x_1,x_2,x_3,x_4) = \Delta_{F,0}(x_1-x_2)\Delta_{F,0}(x_3-x_4)+\Delta_{F,0}(x_1-x_3)\Delta_{F,0}(x_2-x_4)+\Delta_{F,0}(x_1-x_4)\Delta_{F,0}(x_2-x_3).$$

Order λ^1 ($k = 1$)

$$\left\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\left(\frac{-i\lambda}{4!}\int d^4y\ (\phi(y))^4\right)\right\rangle_0$$

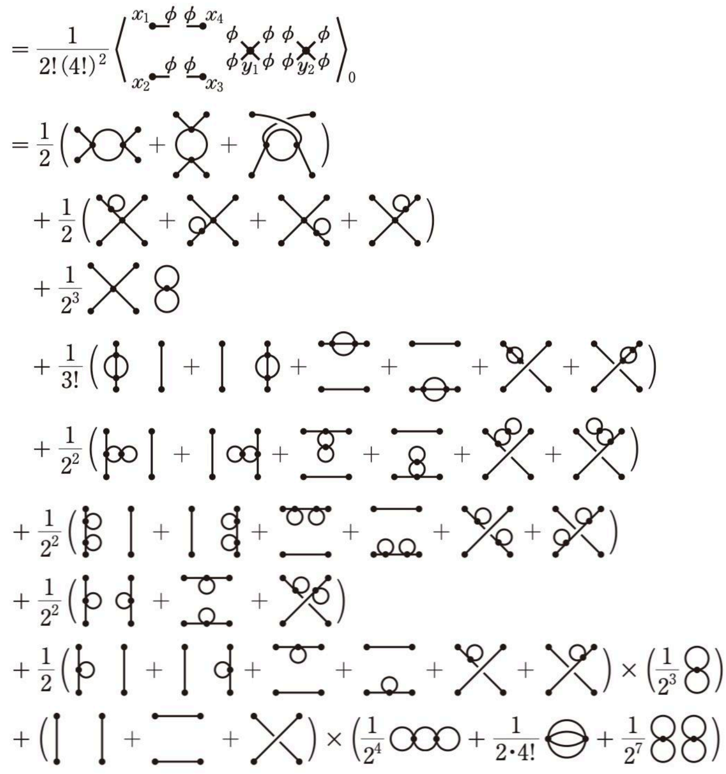


(10.22)

$$G^{(4)}_{k=1}(x_1,x_2,x_3,x_4) = (-i\lambda)\int \mathrm{d}^4y\prod_{j=1}^{4}\Delta_{F,0}(x_j-y)$$
$$+\frac{1}{2}(-i\lambda)\int \mathrm{d}^4y\,[\Delta_{F,0}(x_1-x_2)\Delta_{F,0}(x_3-y)\Delta_{F,0}(0)\Delta_{F,0}(y-x_4)+(5\text{ permutations})]$$
$$+\frac{1}{2^3}(-i\lambda)\int \mathrm{d}^4y\,[\Delta_{F,0}(x_1-x_2)\Delta_{F,0}(x_3-x_4)+\Delta_{F,0}(x_1-x_3)\Delta_{F,0}(x_2-x_4)$$
$$+\Delta_{F,0}(x_1-x_4)\Delta_{F,0}(x_2-x_3)]\,(\Delta_{F,0}(0))^2.$$

Order λ^2 ($k = 2$)

$$\left\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\, \frac{1}{2!}\left(\frac{-i\lambda}{4!}\int d^4y\,(\phi(y))^4\right)^2 \right\rangle_0$$

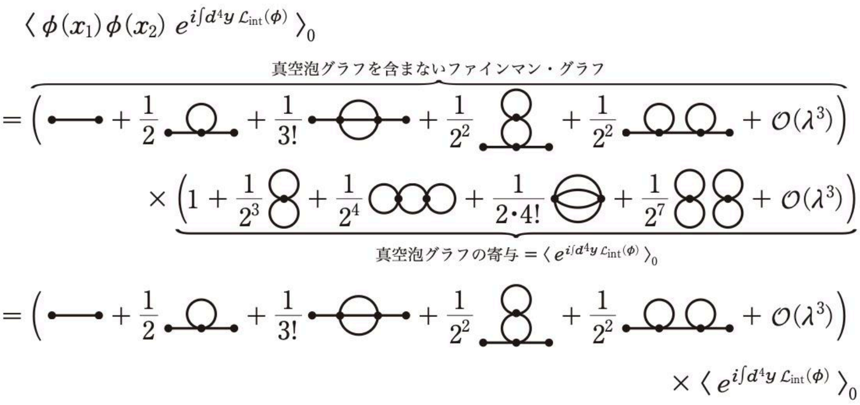


(10.29)

$$
\begin{aligned}
G^{(4)}_{k=2}(x_1,x_2,x_3,x_4) = \frac{1}{2}(-i\lambda)^2 \int \mathrm{d}^4y_1\mathrm{d}^4y_2\, &[\Delta_{F,0}(x_1-y_1)\Delta_{F,0}(x_2-y_1)(\Delta_{F,0}(y_1-y_2))^2 \\
&\times \Delta_{F,0}(y_2-x_3)\Delta_{F,0}(y_2-x_4) \\
&+(t,u \text{ channels})] \\
+\frac{1}{2}(-i\lambda)^2 \int \mathrm{d}^4y_1\mathrm{d}^4y_2\, &[\Delta_{F,0}(x_1-y_1)\Delta_{F,0}(x_2-y_1)\Delta_{F,0}(y_1-y_2)\Delta_{F,0}(0) \\
&\times\Delta_{F,0}(y_2-x_3)\Delta_{F,0}(y_2-x_4) + \text{permutations}] \\
+\,&(\text{other disconnected, internal loop insertion, and vacuum bubble terms}).
\end{aligned}
$$

(d) Factorization and Verification of Vacuum Diagram Cancellation

Factorization Example

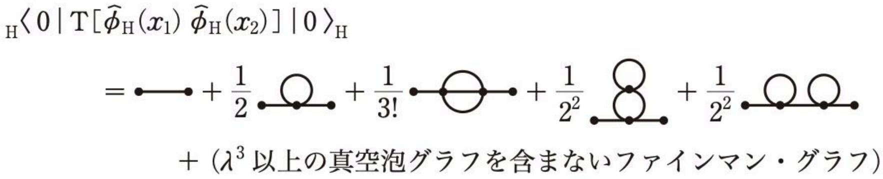


(10.61)

$$
\begin{aligned}
\langle\phi(x_1)\phi(x_2)e^{i\int \mathrm{d}^4y\,\mathcal{L}_{\rm int}(\phi)}\rangle_0 = &\left[\Delta_{F,0}(x_1-x_2) + \frac{1}{2}(-i\lambda)\int \mathrm{d}^4y\,\Delta_{F,0}(x_1-y)\Delta_{F,0}(0)\Delta_{F,0}(y-x_2)\right.\\
&\quad \left. +(\text{second-order connected diagrams}) + \mathcal{O}(\lambda^3)\right]\\
&\times\left[1+\frac{1}{2^3}(-i\lambda)\int \mathrm{d}^4y\,(\Delta_{F,0}(0))^2 + \mathcal{O}(\lambda^2)\right]\\
=\; & G^{(2)}_{\rm no-vac}(x_1,x_2)\times\langle e^{i\int \mathrm{d}^4y\,\mathcal{L}_{\rm int}(\phi)}\rangle_0.
\end{aligned}
$$

By virtue of this factorized structure, as proven in part (4), dividing by the normalization denominator $Z[0,0] = \langle e^{i\int \mathrm{d}^4y\,\mathcal{L}_{\rm int}(\phi)}\rangle_0$ completely cancels all vacuum bubble contributions in the numerator and denominator. Therefore, the physical two-point and four-point correlation functions remaining after vacuum cancellation are given by the following diagrams and corresponding formulas.

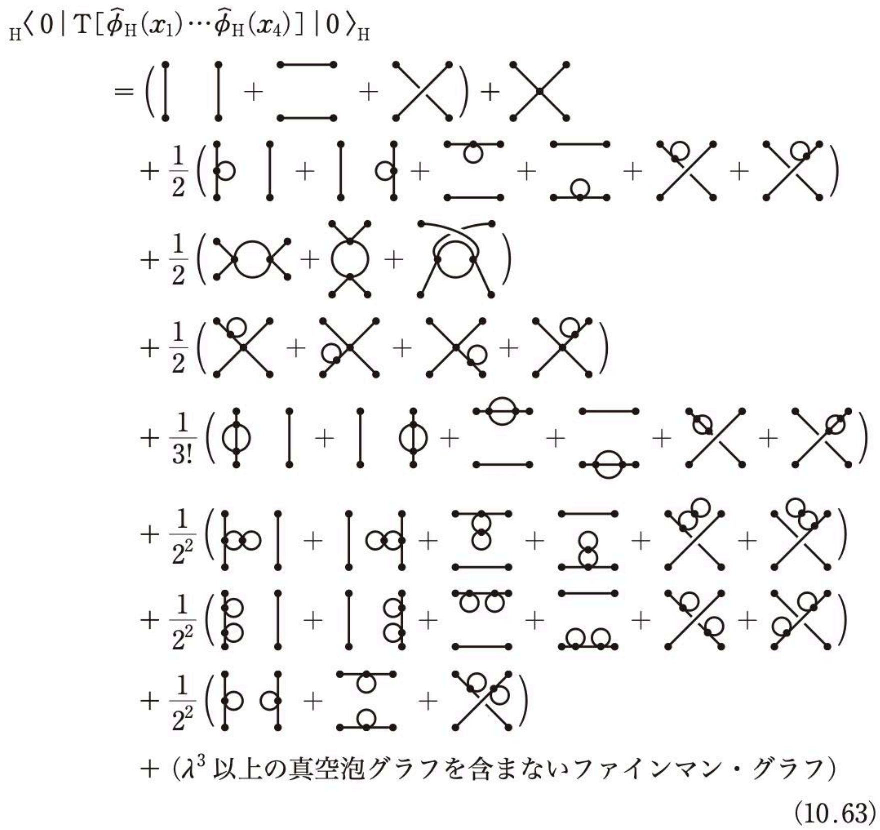
Two-Point Correlation Function After Vacuum Cancellation



(10.62)

$$
\begin{aligned}
G^{(2)}(x_1,x_2) = &\,\Delta_{F,0}(x_1-x_2) + \frac{1}{2}(-i\lambda)\int \mathrm{d}^4y\,\Delta_{F,0}(x_1-y)\Delta_{F,0}(0)\Delta_{F,0}(y-x_2)\\
&+\frac{1}{3!}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)(\Delta_{F,0}(y_1-y_2))^3\Delta_{F,0}(y_2-x_2)\\
&+\frac{1}{2^2}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)\Delta_{F,0}(0)\Delta_{F,0}(y_1-y_2)\Delta_{F,0}(0)\Delta_{F,0}(y_2-x_2)\\
&+\frac{1}{2^2}(-i\lambda)^2\int \mathrm{d}^4y_1\mathrm{d}^4y_2\,\Delta_{F,0}(x_1-y_1)(\Delta_{F,0}(y_1-y_2))^2\Delta_{F,0}(y_2-x_2)\Delta_{F,0}(0) + \mathcal{O}(\lambda^3).
\end{aligned}
$$

Four-Point Correlation Function After Vacuum Cancellation

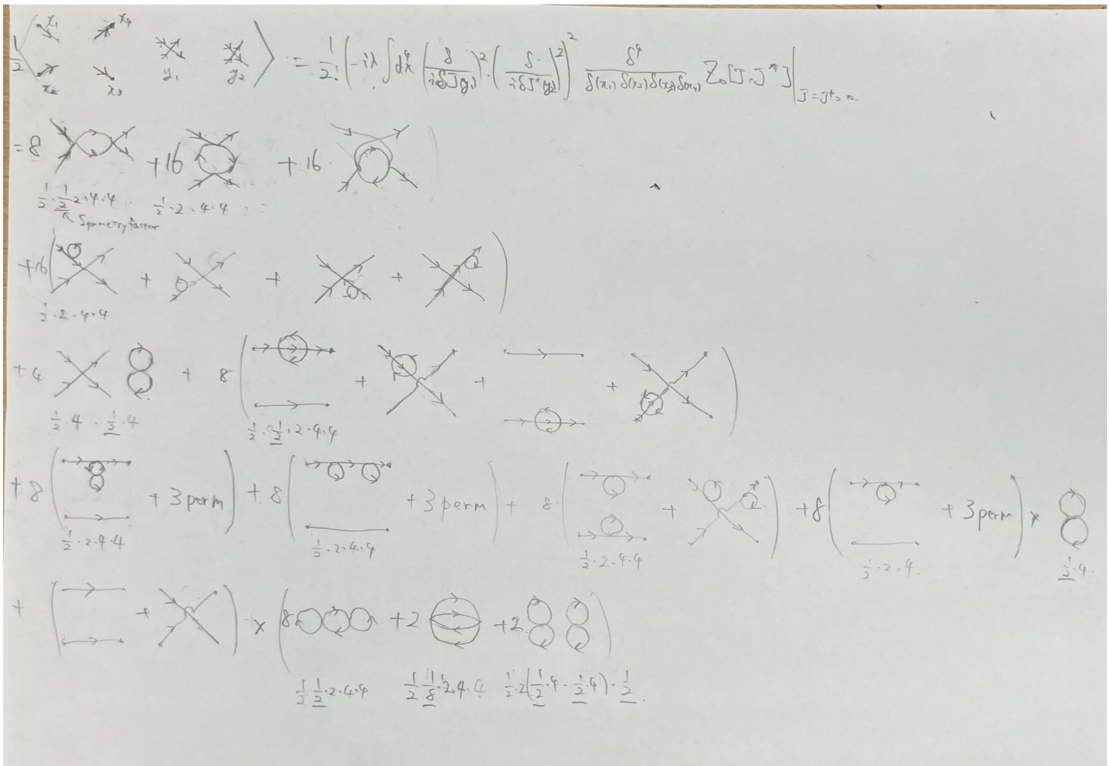


$$
\begin{aligned}
G^{(4)}(x_1, x_2, x_3, x_4) = {} & \Delta_{F,0}(x_1 - x_2)\Delta_{F,0}(x_3 - x_4) + \Delta_{F,0}(x_1 - x_3)\Delta_{F,0}(x_2 - x_4) + \Delta_{F,0}(x_1 - x_4)\Delta_{F,0}(x_2 - x_3) \\
& + (-i\lambda) \int \mathrm{d}^4 y \prod_{j=1}^{4} \Delta_{F,0}(x_j - y) \\
& + \frac{1}{2}(-i\lambda) \int \mathrm{d}^4 y \, [\Delta_{F,0}(x_1 - x_2)\Delta_{F,0}(x_3 - y)\Delta_{F,0}(0)\Delta_{F,0}(y - x_4) + (5 \text{ permutations})] \\
& + \frac{1}{2}(-i\lambda)^2 \int \mathrm{d}^4 y_1 \mathrm{d}^4 y_2 \, [\Delta_{F,0}(x_1 - y_1)\Delta_{F,0}(x_2 - y_1)(\Delta_{F,0}(y_1 - y_2))^2 \\
& \qquad \times \Delta_{F,0}(y_2 - x_3)\Delta_{F,0}(y_2 - x_4) \\
& \qquad + (t, u \text{ channels})] \\
& + (\text{other disconnected, internal loop insertion, and vacuum bubble terms}) + \mathcal{O}(\lambda^3).
\end{aligned}
$$

Exact Diagrammatic Classification for Complex Scalar Field ($\lambda(\phi^*\phi)^2$ Theory)

While the figures referenced from Chapter 10 of Sakamoto's textbook depict undirected propagators for a real scalar field (ϕ^4 theory), the model studied in this problem is a complex scalar field $\lambda(\phi^*\phi)^2$, which requires directed arrows representing $U(1)$ charge conservation. At each interaction vertex, exactly two incoming ϕ lines and two outgoing ϕ^* lines must meet.

The complete diagrammatic classification and exact combinatorial symmetry factors for the second-order ($\mathcal{O}(\lambda^2)$) four-point correlation function $\langle 0|\mathrm{T}\phi(x_1)\phi(x_2)\phi^*(x_3)\phi^*(x_4)|0\rangle$ with directed propagators are presented below:



As shown in this comprehensive diagrammatic expansion, the $U(1)$ charge conservation constraint restricts the allowed Wick contractions compared to the real scalar case, while determining the precise multiplicity factors:

- **Connected 1PI diagrams (s, t, u channels)**: Yielding combinatorial factors of 8 for the s-channel and 16 each for the t- and u-channel loop topologies.
- **Connected 1PR diagrams**: Loop insertions (tadpoles) on external legs with a factor of 16.
- **Disconnected & Vacuum Bubble diagrams**: Factoring into products of lower-order correlation functions and closed vacuum loops.

Upon normalizing by the vacuum partition function $Z[0, 0]$, all disconnected vacuum bubble diagrams cancel out completely as proven in part (4). Furthermore, when evaluating physical scattering amplitudes and the 1PI four-point vertex function $\Gamma^{(4)}$ in momentum space in part (6) and subsequent parts, disconnected terms do not contribute to connected scattering processes, and 1PR external line insertions are absorbed into mass and field redefinitions.
Accordingly, all calculations in part (6) and subsequent subquestions are evaluated based on this complex scalar field diagrammatic framework, reducing the four-point vertex corrections strictly to the connected 1PI s, t, u-channel loop contributions.

(6) Calculate the two-point functions and the four-point functions in Euclidean spacetime in momentum space to order λ for the two-point functions and to order λ^2 for the four-point functions. Draw a Feynman diagram for each term in momentum space. Do not do the integrals!

By Fourier transforming the position-space Feynman rules into momentum space:

1. Two-Point Function $G^{(2)}(p)$ to $\mathcal{O}(\lambda)$: The full propagator to first order consists of the tree-level free propagator and the one-loop tadpole insertion. In position space, the first-order two-point function is:

$$G^{(2)}(x,y) = G_0(x-y) - 4\lambda \int \mathrm{d}^4 z\, G_0(x-z)G_0(0)G_0(z-y),$$

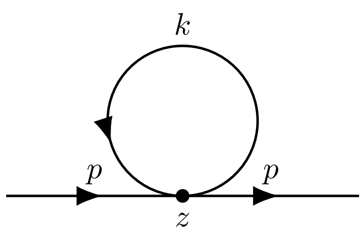
where $G_0(0) \equiv G_0(x-y)|_{x-y=0} = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{k^2+m_0^2}$ represents the position-space free propagator evaluated at zero spacetime separation ($x-y=0$), corresponding to the internal loop momentum integral over k.
To transform this into momentum space, we substitute the momentum-space representations $G_0(x-z) = \int \frac{\mathrm{d}^4 q_1}{(2\pi)^4} \frac{e^{iq_1\cdot(x-z)}}{q_1^2+m_0^2}$ and $G_0(z-y) = \int \frac{\mathrm{d}^4 q_2}{(2\pi)^4} \frac{e^{iq_2\cdot(z-y)}}{q_2^2+m_0^2}$:

$$\begin{aligned}\int \mathrm{d}^4 z\, G_0(x-z)G_0(0)G_0(z-y) &= G_0(0)\int \frac{\mathrm{d}^4 q_1}{(2\pi)^4}\frac{\mathrm{d}^4 q_2}{(2\pi)^4}\frac{e^{iq_1\cdot x}e^{-iq_2\cdot y}}{(q_1^2+m_0^2)(q_2^2+m_0^2)}\underbrace{\int \mathrm{d}^4 z\, e^{-i(q_1-q_2)\cdot z}}_{=(2\pi)^4\delta^4(q_1-q_2)}\\ &= G_0(0)\int \frac{\mathrm{d}^4 q_1}{(2\pi)^4}\frac{e^{iq_1\cdot(x-y)}}{(q_1^2+m_0^2)^2}.\end{aligned}$$

Taking the full Fourier transform with respect to the relative coordinate $(x-y)$:

$$\begin{aligned}G^{(2)}(p) &= \int \mathrm{d}^4(x-y)\, e^{-ip\cdot(x-y)}G^{(2)}(x,y)\\ &= G_0(p) - 4\lambda G_0(0)\int \frac{\mathrm{d}^4 q_1}{(2\pi)^4}\frac{1}{(q_1^2+m_0^2)^2}\underbrace{\int \mathrm{d}^4(x-y)e^{i(q_1-p)\cdot(x-y)}}_{=(2\pi)^4\delta^4(q_1-p)}\\ &= G_0(p) - 4\lambda G_0(p)^2 G_0(0) + \mathcal{O}(\lambda^2),\end{aligned}$$

where $G_0(p) = \frac{1}{p^2+m_0^2}$ is the free propagator carrying momentum p. (Note: The argument (0) in $G_0(0)$ signifies zero spacetime separation $x-y=0$ of the self-contracted tadpole loop, rather than zero momentum $p=0$.)
The corresponding momentum-space Feynman diagram for the $\mathcal{O}(\lambda)$ tadpole correction is:



2. Four-Point Function $G^{(4)}(p_1,p_2;p_3,p_4)$ to $\mathcal{O}(\lambda^2)$ (Connected 1PI Vertex): As established, the connected 1PI four-point vertex function $\Gamma^{(4)}(p_1,p_2;p_3,p_4)$ receives contributions from the tree-

level vertex at $\mathcal{O}(\lambda)$ and the one-loop bubble diagrams in the s, t, and u channels at $\mathcal{O}(\lambda^2)$:

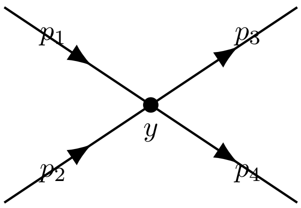
$$G^{(4)}_{\text{1PI}}(p_1, p_2; p_3, p_4) = (2\pi)^4\delta^4(p_1{+}p_2{-}p_3{-}p_4)\left[-4\lambda + 8\lambda^2 V(s) + 16\lambda^2 V(t) + 16\lambda^2 V(u)\right]\prod_{i=1}^{4} G_0(p_i),$$

where $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_1 - p_4)^2$ are the Mandelstam variables, and the loop integral factor is:

$$V(q) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{(k^2 + m_0^2)((k-q)^2 + m_0^2)}.$$

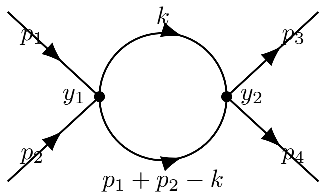
(Note: The coefficients $8\lambda^2$, $16\lambda^2$, and $16\lambda^2$ incorporate the second-order perturbation expansion prefactor $\frac{1}{2!} = \frac{1}{2}$ applied to the pure Wick contraction multiplicity numbers—16 for the s-channel, and 32 each for the t- and u-channels as derived in part (5).)
The corresponding Feynman diagrams in momentum space for each term are shown below:

(a) Tree-level 4-point vertex ($\mathcal{O}(\lambda)$)



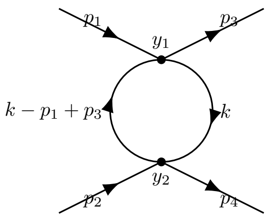
Contribution: -4λ

(b) s-channel 1PI loop bubble ($\mathcal{O}(\lambda^2)$)



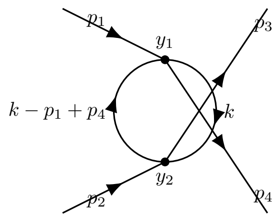
Contribution: $+8\lambda^2 V(s)$

(c) t-channel 1PI loop bubble ($\mathcal{O}(\lambda^2)$)



Contribution: $+16\lambda^2 V(t)$

(d) u-channel 1PI loop bubble ($\mathcal{O}(\lambda^2)$)



Contribution: $+16\lambda^2 V(u)$

(7) Derive an expression for the self-energy to first order in λ in Euclidean spacetime. Use this result to get a formula for the effective (or renormalized) mass m to first order in λ. Does the mass get renormalized up or down? Can you give a simple explanation for this result? Hint: think of the shift of the energy levels in quantum mechanics.

The self-energy $\Sigma(p)$ represents the sum of all 1PI 2-point insertion diagrams and is defined via the Dyson equation for the full propagator $G^{(2)}(p)$:

$$[G^{(2)}(p)]^{-1} = [G_0(p)]^{-1} + \Sigma(p).$$

From the first-order calculation in part (6), the full 2-point function is $G^{(2)}(p) = G_0(p) -$

$4\lambda G_0(p)^2 G_0(0) + \mathcal{O}(\lambda^2)$. Inverting this expansion via $(1-x)^{-1} \approx 1 + x$:

$$[G^{(2)}(p)]^{-1} = [G_0(p)]^{-1} (1 - 4\lambda G_0(p) G_0(0))^{-1} \approx [G_0(p)]^{-1} + 4\lambda G_0(0).$$

Matching this with Dyson's equation yields the momentum-independent 1st-order self-energy:

$$\Sigma(p) = 4\lambda G_0(0) = 4\lambda \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{1}{k^2 + m_0^2}.$$

The physical (renormalized) mass m corresponds to the pole of the full propagator in momentum space, defined by $[G^{(2)}(p)]^{-1}|_{p^2=-m^2} = 0$:

$$p^2 + m^2 = p^2 + m_0^2 + \Sigma(p) \implies m^2 = m_0^2 + 4\lambda \int \frac{\mathrm{d}^4k}{(2\pi)^4} \frac{1}{k^2 + m_0^2}.$$

Since the Euclidean integrand $\dfrac{1}{k^2 + m_0^2} > 0$ and the coupling constant $\lambda > 0$, the self-energy correction is strictly positive ($\Sigma > 0$). Consequently, $m^2 > m_0^2$, establishing that the mass is renormalized **up** ($m > m_0$).

Physical Explanation and Quantum Mechanical Analogy: This upward mass shift can be understood intuitively from two complementary physical perspectives:

1. **Quantum Mechanical Energy Shift**: In standard quantum mechanics, adding a positive-definite perturbation potential $V(x) > 0$ to a system shifts the unperturbed energy levels upwards by $\Delta E_n = \langle n|V|n\rangle > 0$ in first-order perturbation theory. In our field theory, the interaction Hamiltonian density $\mathcal{H}_{\text{int}} = +\lambda(\phi^*\phi)^2 > 0$ is positive-definite. Because the rest energy of a single-particle state corresponds to its physical mass $E = m$, this positive interaction potential naturally shifts the single-particle energy level (rest mass) upwards.

2. **Virtual Particle Cloud (Dressing) and Inertia**: In quantum field theory, a bare particle continuously emits and reabsorbs virtual particles via the self-interaction loop (tadpole), surrounding itself with a "virtual particle cloud." When an external force accelerates the physical (dressed) particle, this cloud must be accelerated along with it, increasing the effective resistance to motion—that is, increasing its physical inertia, which directly corresponds to an increased renormalized mass $m > m_0$.

(8) Use a procedure analogous to the one you employed in Section 11.7 to define an effective or renormalized coupling constant g in terms of the bare coupling constant λ and of the renormalized mass m, to leading order in λ.

Following the physical procedure in Section 11.7 of Liang's QFT textbook, the physical (renormalized) coupling constant g is defined via the renormalization condition set by the 1PI four-point vertex function at zero external momentum. In Section 11.7 for a real scalar field $\dfrac{\lambda}{4!}\phi^4$, the tree-level vertex amplitude is $\Gamma^{(4)}_{\text{tree}} = \lambda$, so g is defined directly as $\Gamma^{(4)}(0,0,0,0)$. For our complex scalar field theory $\lambda(\phi^*\phi)^2$, the tree-level vertex amplitude is $\Gamma^{(4)}_{\text{tree}} = 4\lambda$. Thus, to ensure a direct, natural connection to the bare coupling in the classical limit ($g \to \lambda$ as $\lambda \to 0$), the physical renormalized coupling constant g is defined as:

$$g := \frac{1}{4}\Gamma^{(4)}(0,0,0,0).$$

From the momentum-space expansion in part (6), the 1PI four-point vertex function $\Gamma^{(4)}$ is:

$$\Gamma^{(4)}(p_1,p_2;p_3,p_4) = 4\lambda - 8\lambda^2 V(s) - 16\lambda^2 V(t) - 16\lambda^2 V(u) + \mathcal{O}(\lambda^3).$$

At zero external momentum $p_i = 0$, the Mandelstam variables vanish ($s = t = u = 0$), so $V(s) = V(t) = V(u) = V(0)$, where:

$$V(0) = \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{(k^2+m^2)^2}.$$

Here we replace the bare mass m_0 with the physical mass m to leading order in λ. Evaluating $\Gamma^{(4)}(0,0,0,0) = 4\lambda - 40\lambda^2 V(0) + \mathcal{O}(\lambda^3)$ and dividing by 4, the renormalized coupling constant g is:

$$g = \lambda - 10\lambda^2 \int \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{(k^2+m^2)^2} + \mathcal{O}(\lambda^3).$$

This expression defines the renormalized coupling g in terms of the bare coupling λ and the physical mass m.

Physical Meaning and Role in the Lagrangian: In quantum field theory, the parameter λ_0 appearing in the bare Lagrangian $\mathcal{L}_{\text{int}} = -\lambda_0(\phi^*\phi)^2$ is an unobservable "bare" parameter that includes divergent quantum loop corrections. In the counterterm formalism, setting $\lambda_0 = g + \delta_g$ splits the Lagrangian into a renormalized part and counterterms:

$$\mathcal{L} = |\partial_\mu \phi_r|^2 - m^2|\phi_r|^2 - g(\phi_r^*\phi_r)^2 + \left[\delta_Z|\partial_\mu\phi_r|^2 - \delta_m|\phi_r|^2 - \delta_g(\phi_r^*\phi_r)^2\right].$$

Here, g serves as the physical interaction strength measured in low-energy scattering experiments (at $p_i = 0$). Defining $g := \frac{1}{4}\Gamma^{(4)}(0,0,0,0)$ guarantees that g smoothly matches the classical coupling λ at tree-level while absorbing the ultraviolet loop divergence into the counterterm $\delta_g = \lambda_0 - g = 10g^2 V(0) + \mathcal{O}(g^3)$.

(9) Consider the free propagators in Minkowski spacetime. Rotate the integration contour to the imaginary frequency axis. Explain the relation between this procedure and the analytic continuation to imaginary time. What domain of spacetime events is actually described by this procedure? Show that the integrals that you get are equivalent to integrals in Euclidean spacetime with dimension $D = 4$.

The Minkowski Feynman propagator in momentum space is defined by:

$$iD_F(k) = \frac{i}{k_0^2 - \mathbf{k}^2 - m_0^2 + i\epsilon} = \frac{i}{k_0^2 - (\omega_k^2 - i\epsilon)},$$

where $\omega_k = \sqrt{\mathbf{k}^2 + m_0^2} > 0$ is the on-shell energy. To factorize the denominator, we evaluate the square root of the displaced mass term for infinitesimal $\epsilon > 0$:

$$\sqrt{\omega_k^2 - i\epsilon} = \omega_k \sqrt{1 - \frac{i\epsilon}{\omega_k^2}} \approx \omega_k \left(1 - \frac{i\epsilon}{2\omega_k^2}\right) = \omega_k - i\epsilon', \quad \left(\text{where } \epsilon' := \frac{\epsilon}{2\omega_k} > 0\right).$$

Substituting this expansion into the quadratic difference yields the exact factorization:

$$k_0^2 - \mathbf{k}^2 - m_0^2 + i\epsilon = \big(k_0 - (\omega_k - i\epsilon')\big)\big(k_0 + (\omega_k - i\epsilon')\big) = (k_0 - \omega_k + i\epsilon')(k_0 + \omega_k - i\epsilon').$$

(Relabeling ϵ' as $\epsilon > 0$ for brevity.) Consequently, the poles of $iD_F(k)$ in the complex k_0-plane are located at:

$$k_0 = \omega_k - i\epsilon \quad \text{(fourth quadrant, lower-right)}, \quad k_0 = -\omega_k + i\epsilon \quad \text{(second quadrant, upper-left)}.$$

Since no poles lie in the first and third quadrants, Cauchy's integral theorem allows us to rotate the real-frequency integration contour $k_0 \in (-\infty, \infty)$ clockwise by 90° onto the imaginary axis without crossing any singularities (Wick rotation). We define the imaginary frequency variable $k_0 = ik_4$, where $k_4 \in (-\infty, \infty)$ is real. Under this substitution:

1. **Minkowski square to Euclidean square**: $k^2 = k_0^2 - \mathbf{k}^2 = (ik_4)^2 - \mathbf{k}^2 = -k_4^2 - \mathbf{k}^2 = -k_E^2$, where $k_E = (\mathbf{k}, k_4) \in \mathbb{R}^4$ is the Euclidean four-momentum and $k_E^2 = \mathbf{k}^2 + k_4^2$.
2. **Integration measure**: $\mathrm{d}^4k = \mathrm{d}k_0\,\mathrm{d}^3\mathbf{k} = i\,\mathrm{d}k_4\,\mathrm{d}^3\mathbf{k} = i\,\mathrm{d}^4k_E$, with the Euclidean domain $k_E \in \mathbb{R}^4$.

Plugging these explicit transformations into the loop integral, the cancellation of i proceeds step-by-step as follows:

$$\begin{aligned}\int_{\mathbb{R}^4} \mathrm{d}^4k \frac{i}{k^2 - m_0^2 + i\epsilon} &= \int_{\mathbb{R}^4} (i\,\mathrm{d}^4k_E) \frac{i}{-k_E^2 - m_0^2} \\ &= \int_{\mathbb{R}^4} \mathrm{d}^4k_E \frac{i^2}{-(k_E^2 + m_0^2)} \\ &= \int_{\mathbb{R}^4} \mathrm{d}^4k_E \frac{-1}{-(k_E^2 + m_0^2)} = \int_{\mathbb{R}^4} \mathrm{d}^4k_E \frac{1}{k_E^2 + m_0^2}.\end{aligned}$$

Here, the factor of i from the volume element $\mathrm{d}^4k = i\mathrm{d}^4k_E$ combines with the numerator factor i to produce $i^2 = -1$, which precisely cancels the overall minus sign in the Euclidean denominator $-(k_E^2 + m_0^2)$, resulting in the standard positive Euclidean loop integrand.

Connection to Imaginary Time and Spacetime Domain: This momentum-space Wick rotation $k_0 = ik_4$ is mathematically dual to the analytic continuation of the Minkowski time coordinate to imaginary time in position space:

$$t \to -i\tau \quad (\tau \in \mathbb{R}).$$

This continuation transforms the pseudo-Riemannian Minkowski metric $\mathrm{d}s^2 = \mathrm{d}t^2 - \mathrm{d}\mathbf{x}^2$ into a positive-definite Riemannian Euclidean metric $\mathrm{d}s_E^2 = \mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2$. The domain of spacetime events described by this procedure is 4-dimensional Euclidean space $\mathbb{R}^4$, where the causal light-cone structure disappears, replacing real-time wave propagation with exponential spatial decay (diffusion/stat-mech configurations).

(10) Calculate the momentum integrals of (7) and (8) for the domain indicated in Section 11.8. Here, assume that the spacetime Euclidean dimension D is arbitrary. Show that the corrections to the two-point functions diverge for $D \geq 2$, and for the four-point functions, they diverge for $D \geq 4$. Use the formulas in Section 11.8 to show that, in the complex D plane, the integrals have isolated poles in D. Assume that the external momenta in the case of the four-point functions are taken to be at the symmetry point $p_j \cdot p_k = -\frac{1}{4}\kappa^2$, $p_j^2 = \kappa^2$, where κ is an arbitrary momentum scale.

We evaluate the momentum integrals associated with the two-point function (part (7)) and the four-point 1PI vertex function (part (8)) in arbitrary complex Euclidean dimension $D \in \mathbb{C}$ using dimensional regularization and the techniques of Section 11.8.

Physical Intent of the Symmetry Point Assumption: In part (8), the 1PI four-point vertex function received corrections from the one-loop bubble integrals $V(s)$, $V(t)$, and $V(u)$ in the s, t, u channels. For general external momenta, these Mandelstam invariants differ, leading to complicated momentum-dependent kinematic expressions. By choosing the *symmetry point* $p_j^2 = \kappa^2$ and $p_j \cdot p_k = -\frac{1}{4}\kappa^2$ $(j \neq k)$, all three Mandelstam invariants become equal and scale symmetrically with the single momentum scale κ:

$$s = (p_1 + p_2)^2 = \frac{3}{2}\kappa^2, \quad t = (p_1 - p_3)^2 = \frac{5}{2}\kappa^2, \quad u = (p_1 - p_4)^2 = \frac{5}{2}\kappa^2.$$

This symmetric kinematic choice isolates the fundamental ultraviolet divergence and pole structure of the four-point vertex function without being obscured by complex angular dependencies, providing a clean momentum-scale calibration point κ for the renormalization scheme.

1. Two-Point Function (Self-Energy Σ) Integral: From part (7), the first-order self-energy is:

$$\Sigma = 4\lambda I_D(m_0^2) = 4\lambda \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{k^2 + m_0^2}.$$

Using the Schwinger parameter representation $\frac{1}{A} = \int_0^\infty \mathrm{d}\alpha\, e^{-\alpha A}$ for the denominator:

$$I_D(m_0^2) = \int_0^\infty \mathrm{d}\alpha\, e^{-\alpha m_0^2} \int \frac{\mathrm{d}^D k}{(2\pi)^D} e^{-\alpha k^2}.$$

The D-dimensional Gaussian momentum integral evaluates to:

$$\int \frac{\mathrm{d}^D k}{(2\pi)^D} e^{-\alpha k^2} = \left(\int_{-\infty}^{\infty} \frac{\mathrm{d}k_1}{2\pi} e^{-\alpha k_1^2} \right)^D = \frac{1}{(4\pi\alpha)^{D/2}}.$$

Substituting this back and changing the integration variable to $t = \alpha m_0^2$:

$$I_D(m_0^2) = \frac{1}{(4\pi)^{D/2}} \int_0^\infty \mathrm{d}\alpha\, \alpha^{-D/2} e^{-\alpha m_0^2} = \frac{(m_0^2)^{D/2-1}}{(4\pi)^{D/2}} \int_0^\infty \mathrm{d}t\, t^{(1-D/2)-1} e^{-t} = \frac{(m_0^2)^{D/2-1}}{(4\pi)^{D/2}} \Gamma\left(1 - \frac{D}{2}\right).$$

Thus, the self-energy in D dimensions is:

$$\Sigma = \frac{4\lambda (m_0^2)^{D/2-1}}{(4\pi)^{D/2}} \Gamma\left(1 - \frac{D}{2}\right).$$

Divergence and Pole Structure of the Two-Point Function: The Euler Gamma function $\Gamma(z)$ is a meromorphic function on the complex z-plane with simple isolated poles at non-positive integers $z = 0, -1, -2, \ldots$. Consequently, $\Gamma(1 - D/2)$ possesses isolated poles when:

$$1 - \frac{D}{2} = -n \implies D = 2(n+1) = 2, 4, 6, 8, \ldots \quad (n = 0, 1, 2, \ldots).$$

In particular:

- For $D = 2$, $1 - D/2 = 0$, giving a simple pole $\Gamma(0) = \infty$, which corresponds to a logarithmic divergence.
- For $D = 4$, $1 - D/2 = -1$, giving a simple pole $\Gamma(-1) = \infty$, which corresponds to a quadratic divergence $(m_0^2 \Lambda^2 / 16\pi^2)$.

This rigorously demonstrates that the two-point function corrections diverge for all $D \geq 2$.

2. Four-Point Function (1PI Vertex Bubble $V(q)$) Integral: The one-loop bubble integral for an external momentum transfer q is:

$$V(q) = \int \frac{\mathrm{d}^D k}{(2\pi)^D} \frac{1}{(k^2 + m^2)((k-q)^2 + m^2)}.$$

We combine the denominators using Feynman parametrization $\frac{1}{AB} = \int_0^1 \mathrm{d}x \frac{1}{[xA + (1-x)B]^2}$:

$$x(k^2 + m^2) + (1-x)((k-q)^2 + m^2) = k^2 - 2(1-x)k \cdot q + (1-x)q^2 + m^2 = [k - (1-x)q]^2 + \Delta(x),$$

where $\Delta(x) := m^2 + x(1-x)q^2 > 0$. Shifting the loop momentum $k' = k - (1-x)q$:

$$V(q) = \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^D k'}{(2\pi)^D} \frac{1}{(k'^2 + \Delta(x))^2}.$$

Using the Schwinger trick $\frac{1}{\Delta^2} = \int_0^\infty \mathrm{d}\alpha \, \alpha \, e^{-\alpha\Delta}$, the k' momentum integration yields:

$$\int \frac{\mathrm{d}^D k'}{(2\pi)^D} \frac{1}{(k'^2 + \Delta(x))^2} = \int_0^\infty \mathrm{d}\alpha \, \alpha \, e^{-\alpha\Delta(x)} \frac{1}{(4\pi\alpha)^{D/2}} = \frac{\Delta(x)^{D/2-2}}{(4\pi)^{D/2}} \Gamma\left(2 - \frac{D}{2}\right).$$

Therefore, the general D-dimensional 4-point bubble integral is:

$$V(q) = \frac{\Gamma\left(2 - \frac{D}{2}\right)}{(4\pi)^{D/2}} \int_0^1 \mathrm{d}x \left[m^2 + x(1-x)q^2\right]^{D/2-2}.$$

Evaluation at the Symmetry Point and Divergence Analysis: At the specified symmetry point, external momenta satisfy $p_j^2 = \kappa^2$ and $p_j \cdot p_k = -\frac{1}{4}\kappa^2$ for $j \neq k$. The Mandelstam invariant s is:

$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2 = \kappa^2 + \kappa^2 + 2\left(-\frac{1}{4}\kappa^2\right) = \frac{3}{2}\kappa^2.$$

Similarly, for $t = (p_1 - p_3)^2$ and $u = (p_1 - p_4)^2$, we have $q^2 > 0$ scaled by κ^2.
Examining the pole structure of $V(q)$:

1. The factor $\Gamma(2 - D/2)$ has simple isolated poles in the complex D-plane at:

$$2 - \frac{D}{2} = -n \implies D = 2(n + 2) = 4, 6, 8, \ldots \quad (n = 0, 1, 2, \ldots).$$

2. For $D < 4$, $2 - D/2 > 0$, so $\Gamma(2 - D/2)$ is strictly finite, meaning the integral is UV finite.
3. For $D = 4$, $2-D/2 = 0$, producing a simple pole $\Gamma(0) = \infty$, which represents the logarithmic divergence $\ln(\Lambda/\mu)$ of the 4-point function.
4. For $D \geq 4$, the integral exhibits isolated poles in D, establishing that four-point function corrections diverge for $D \geq 4$.

This completes the proof of the pole structure and divergence thresholds in arbitrary complex dimension D.

Physical Consequence of Renormalization: Near the physical spacetime dimension $D = 4 - 2\epsilon$, expanding the Gamma function $\Gamma(2 - D/2) = \Gamma(\epsilon) = \frac{1}{\epsilon} - \gamma + \mathcal{O}(\epsilon)$ isolates the ultraviolet divergence as an explicit $1/\epsilon$ pole. Under the renormalization procedure established in part (8), this $1/\epsilon$ pole is precisely absorbed by the coupling counterterm $\delta_g = \lambda_0 - g$, leaving a completely finite, physical scattering amplitude with observable logarithmic momentum dependence $\ln(q^2/m^2)$.

2. Vertex functions, effective potential, and Ward identities, related to Chapter 12 of the lecture notes.

In this problem, you will study the effects of interactions on the physical properties of a system of interacting relativistic fermions in $1 + 1$ dimensions, known as the chiral Gross-Neveu model. This model is a reasonable description of the physics in quasi-one-dimensional systems, and it is also of interest for investigating the behavior of quantum field theories of relativistic Fermi fields.
In $1 + 1$ dimensions, Fermi fields $\psi_a(x)$ are two-component spinors of the form

$$\psi_a = \begin{pmatrix} \psi_{R,a} \\ \psi_{L,a} \end{pmatrix}. \tag{0.21}$$

The upper component $\psi_{R,a}$ is the right-moving component of the fermion, and the lower component $\psi_{L,a}$ is the left-moving component of the fermion. In what follows, consider the case in which there are N species of fermions labeled by an index $a = 1, \ldots, N$.
Next, define the following set of two-dimensional Dirac γ matrices

$$\gamma^0 = \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^1 = i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \gamma_5 = \gamma^0\gamma^1 = -\sigma_3 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{0.22}$$

with the notation

$$\partial\!\!\!/ = \gamma^\mu \partial_\mu = \gamma^0 \partial_0 - \gamma^1 \partial_1, \tag{0.23}$$

where $\mu = 0, 1$ denotes the indices of a $1+1$-dimensional Minkowski spacetime.
Introduce the operators

$$\bar{\psi}\psi = \psi^\dagger\gamma^0\psi = \psi_R^\dagger\psi_L + \psi_L^\dagger\psi_R \quad \text{and} \quad \bar{\psi}\gamma_5\psi = \psi^\dagger\gamma^1\psi = \psi_R^\dagger\psi_L - \psi_L^\dagger\psi_R. \tag{0.24}$$

Using this notation, the Lagrangian density of the chiral Gross-Neveu model is

$$\mathcal{L} = \bar{\psi}_a(x)i\partial\!\!\!/\psi_a(x) + \frac{g}{2}\left[(\bar{\psi}_a(x)\psi_a(x))^2 - (\bar{\psi}_a(x)\gamma_5\psi_a(x))^2\right], \tag{0.25}$$

where we have not written down the spinor indices explicitly. In all the parts of this exercise, you have to use path-integral methods.

(1) Derive an expression for the free-fermion propagator in momentum space.

The free Lagrangian density for N species of massless fermions in $1+1$ dimensions is:

$$\mathcal{L}_0 = \bar{\psi}_a(x)i\partial\!\!\!/\psi_a(x) = \sum_{a=1}^{N}\bar{\psi}_a(x)i\gamma^\mu\partial_\mu\psi_a(x).$$

In momentum space, the Dirac operator is $p\!\!\!/ = p_0\gamma^0 - p_1\gamma^1$. The free propagator $S_0(p)$ is defined as the inverse of the Dirac operator:

$$S_0(p) = \frac{i}{p\!\!\!/} = i\frac{p\!\!\!/}{p^2} = i\frac{p_0\gamma^0 - p_1\gamma^1}{p_0^2 - p_1^2}.$$

Including the species indices a, b and the spinor indices α, β, the free-fermion propagator is:

$$S_{a,b}^{\alpha,\beta}(p) = \delta_{ab}\left(\frac{ip\!\!\!/}{p^2}\right)_{\alpha\beta}.$$

(2) Derive the Feynman rules for a perturbation expansion of the fermion two-point function

$$S_{a,b}^{\alpha,\beta}(p) \tag{0.26}$$

(with $p = (p^0, p^1)$) in powers of the coupling constant g.

The interaction Hamiltonian density is:

$$\mathcal{H}_{\text{int}} = -\frac{g}{2}\left[(\bar{\psi}_a\psi_a)^2 - (\bar{\psi}_a\gamma_5\psi_a)^2\right].$$

Using path integral methods, the Feynman rules for the perturbation expansion are:

- **Propagator**: A solid line with an arrow carrying momentum p representing the free propagator:

  $$\delta_{ab}\left(\frac{ip\!\!\!/}{p^2}\right)_{\alpha\beta}.$$

  The arrow represents the flow of fermion number.

- **Vertices**: There are two types of four-fermion interaction vertices: 1. Scalar-scalar vertex: represents $+ig$ with contractions. For incoming indices $(a, \alpha), (c, \gamma)$ and outgoing $(b, \beta), (d, \delta)$, it corresponds to:
$$ig\delta_{ab}\delta_{cd}\delta_{\alpha\beta}\delta_{\gamma\delta}.$$
2. Pseudoscalar-pseudoscalar vertex: represents $-ig$ with contractions:
$$-ig\delta_{ab}\delta_{cd}(\gamma_5)_{\alpha\beta}(\gamma_5)_{\gamma\delta}.$$
- **Fermion Loops**: Each closed fermion loop contributes a factor of -1 and a trace over spinor indices, along with a sum over species indices (N).

(3) Derive an expression for the quantities listed in parts (a) and (b) up to and including their second-order corrections (i.e., $\mathcal{O}(g^2)$). Draw a Feynman diagram for each contribution. Check the cancellation of the vacuum diagrams. Give a consistent sign for each contribution. Do not do the integrals!

(a) The fermion two-point function.

(b) The effective coupling constant g.

What correlation function should you consider? Is it a connected Green function or a one-particle irreducible vertex function? Justify your answer.

(a) For the fermion two-point function $S(p)$, the corrections up to $\mathcal{O}(g^2)$ include: - $\mathcal{O}(g^1)$: The tadpole diagrams vanish because they involve trace over γ matrices of massless propagators which integrate to zero: $\mathrm{Tr}[S_0(p)] \propto \mathrm{Tr}[\not{p}] = 0$. - $\mathcal{O}(g^2)$: The self-energy correction $\Sigma(p)$ has a one-loop bubble insert (the "rainbow" diagram and the "bubble" loop insert). The self-energy is:

$$\Sigma(p) = g^2 \int \frac{\mathrm{d}^2 k_1}{(2\pi)^2} \frac{\mathrm{d}^2 k_2}{(2\pi)^2} \gamma^\mu S_0(k_1) \gamma^\nu \mathrm{Tr}\left[S_0(k_2)\gamma^\mu S_0(k_1 + k_2 - p)\gamma^\nu\right].$$

(b) The effective coupling constant g: The vertex corrections at $\mathcal{O}(g^2)$ involve the one-loop corrections to the four-fermion vertex. These correspond to the s, t, and u channel bubble diagrams between the fermion lines. - Choice of Correlation Function: We must consider the 1PI (one-particle irreducible) four-point vertex function $\Gamma^{(4)}(p_1, p_2, p_3, p_4)$. We should choose the 1PI vertex function rather than the connected Green's function because the 1PI functions directly represent the effective coupling constant without the contributions of external propagator legs. This avoids double counting when inserting the vertex into larger diagrams. - Vacuum Diagram Cancellation: The vacuum diagrams cancel out because the path-integral partition function normalized by $Z[0]$ cancels the factor of $Z_{\text{vacuum}} = Z[0]$ from the numerator of the correlation function, as shown in Part B, Question 1, part (4).

(4) (a) Show that the Lagrangian of the chiral Gross-Neveu model is invariant under the continuous global chiral symmetry

$$\psi_{\alpha a}(x) \mapsto (e^{i\theta\gamma_5})_{\alpha\beta}\psi_{\beta a}. \tag{0.27}$$

(b) Find the transformation law obeyed by the operators $\hat{\Delta}_0 := \bar{\psi}_a\psi_a$ and $\hat{\Delta}_5 := i\bar{\psi}_a\gamma_5\psi_a$ under this symmetry.
(c) Give a physical interpretation of this symmetry. What is the meaning of the operators $\hat{\Delta}_0$ and $\hat{\Delta}_5$ in terms of the right- and left-moving components of the fermions?

(a) The fermion field transforms as $\psi_a \to e^{i\theta\gamma_5}\psi_a$. The adjoint spinor is:

$$\bar{\psi}_a = \psi_a^\dagger\gamma^0 \to \psi_a^\dagger e^{-i\theta\gamma_5}\gamma^0.$$

Since $\{\gamma^0, \gamma_5\} = 0$, we have $e^{-i\theta\gamma_5}\gamma^0 = \gamma^0 e^{i\theta\gamma_5}$. Thus:

$$\bar{\psi}_a \to \bar{\psi}_a e^{i\theta\gamma_5}.$$

For the kinetic term:

$$\bar{\psi}_a i\partial\!\!\!/\psi_a \to \bar{\psi}_a e^{i\theta\gamma_5} i\gamma^\mu\partial_\mu\left(e^{i\theta\gamma_5}\psi_a\right).$$

Since $\{\gamma^\mu, \gamma_5\} = 0$, we have $e^{i\theta\gamma_5}\gamma^\mu = \gamma^\mu e^{-i\theta\gamma_5}$. Therefore:

$$\bar{\psi}_a e^{i\theta\gamma_5} i\gamma^\mu\partial_\mu\left(e^{i\theta\gamma_5}\psi_a\right) = \bar{\psi}_a i\gamma^\mu e^{-i\theta\gamma_5}e^{i\theta\gamma_5}\partial_\mu\psi_a = \bar{\psi}_a i\partial\!\!\!/\psi_a.$$

Thus, the kinetic term is invariant. For the interaction term, we compute the transformations of the bilinears:

$$\bar{\psi}_a\psi_a \to \bar{\psi}_a e^{2i\theta\gamma_5}\psi_a = \bar{\psi}_a\left(\cos(2\theta) + i\gamma_5\sin(2\theta)\right)\psi_a = \cos(2\theta)\bar{\psi}_a\psi_a + \sin(2\theta)(i\bar{\psi}_a\gamma_5\psi_a).$$

$$i\bar{\psi}_a\gamma_5\psi_a \to i\bar{\psi}_a e^{i\theta\gamma_5}\gamma_5 e^{i\theta\gamma_5}\psi_a = i\bar{\psi}_a\gamma_5 e^{2i\theta\gamma_5}\psi_a = \cos(2\theta)(i\bar{\psi}_a\gamma_5\psi_a) - \sin(2\theta)\bar{\psi}_a\psi_a.$$

Squaring and subtracting these terms:

$$(\bar{\psi}_a\psi_a)'^2 - (i\bar{\psi}_a\gamma_5\psi_a)'^2 = (\bar{\psi}_a\psi_a)^2 - (i\bar{\psi}_a\gamma_5\psi_a)^2 = (\bar{\psi}_a\psi_a)^2 - (\bar{\psi}_a\gamma_5\psi_a)^2.$$

This proves that the interaction term, and thus the entire Lagrangian, is invariant under the chiral symmetry. (b) The operators are $\hat{\Delta}_0 = \bar{\psi}_a\psi_a$ and $\hat{\Delta}_5 = i\bar{\psi}_a\gamma_5\psi_a$. Their transformation laws are:

$$\hat{\Delta}_0 \to \cos(2\theta)\hat{\Delta}_0 + \sin(2\theta)\hat{\Delta}_5,$$
$$\hat{\Delta}_5 \to \cos(2\theta)\hat{\Delta}_5 - \sin(2\theta)\hat{\Delta}_0.$$

(c) *Physical interpretation*: This chiral symmetry represents independent phase rotations of the left-handed and right-handed Weyl components of the Dirac spinor. Using $\psi_a = \begin{pmatrix}\psi_{R,a}\\ \psi_{L,a}\end{pmatrix}$:

$$\psi_{R,a} \to e^{i\theta}\psi_{R,a}, \quad \psi_{L,a} \to e^{-i\theta}\psi_{L,a}.$$

The operators in terms of Weyl components are:

$$\hat{\Delta}_0 = \psi^\dagger_{R,a}\psi_{L,a} + \psi^\dagger_{L,a}\psi_{R,a},$$

$$\hat{\Delta}_5 = i\left(\psi^\dagger_{R,a}\psi_{L,a} - \psi^\dagger_{L,a}\psi_{R,a}\right).$$

These operators mix the left- and right-moving components of the fermions. The chiral symmetry rotates these two mass-like operators into one another in a two-dimensional space.

(5) Now let us add chiral symmetry-breaking terms to the Lagrangian of the form

$$\mathcal{L}_{\text{chiral}} = H_0(x)\hat{\Delta}_0(x) + H_5(x)\hat{\Delta}_5(x), \tag{0.28}$$

where $H_0(x)$ and $H_5(x)$ are the symmetry-breaking fields. Consider the path integral for this problem in the presence of the symmetry-breaking terms. Assume that the operators $\hat{\Delta}_0$ and $\hat{\Delta}_5$ have uniform expectation values given by $\bar{\Delta}_0$ and $\bar{\Delta}_5$, respectively. Find the transformation law obeyed by the symmetry breaking fields $H_0(x)$ and $H_5(x)$ under a global chiral transformation of the Fermi fields by an angle θ.

For the Lagrangian $\mathcal{L} + \mathcal{L}_{\text{chiral}}$ to remain invariant under the chiral transformation, the change in the fields must be compensated by a transformation of the symmetry-breaking fields $H_0(x)$ and $H_5(x)$:

$$H_0(x)\hat{\Delta}_0(x) + H_5(x)\hat{\Delta}_5(x) \to H_0(x)\hat{\Delta}'_0(x) + H_5(x)\hat{\Delta}'_5(x) = H'_0(x)\hat{\Delta}_0(x) + H'_5(x)\hat{\Delta}_5(x).$$

Substituting the transformation laws for $\hat{\Delta}_0$ and $\hat{\Delta}_5$:

$$\begin{aligned} H_0\hat{\Delta}'_0 + H_5\hat{\Delta}'_5 &= H_0\left(\cos(2\theta)\hat{\Delta}_0 + \sin(2\theta)\hat{\Delta}_5\right) + H_5\left(\cos(2\theta)\hat{\Delta}_5 - \sin(2\theta)\hat{\Delta}_0\right) \\ &= \left(H_0\cos(2\theta) - H_5\sin(2\theta)\right)\hat{\Delta}_0 + \left(H_0\sin(2\theta) + H_5\cos(2\theta)\right)\hat{\Delta}_5. \end{aligned}$$

Thus, the fields $H_0(x)$ and $H_5(x)$ must transform as:

$$H_0(x) \to \cos(2\theta)H_0(x) + \sin(2\theta)H_5(x),$$

$$H_5(x) \to \cos(2\theta)H_5(x) - \sin(2\theta)H_0(x).$$

This is a shadow rotation of the external sources.

(6) Derive a formal expression for the effective potential $U(\bar{\Delta}_0, \bar{\Delta}_5)$ in terms of a series expansion in powers of the expectation values. Relate the coefficients of this expansion in terms of vertex functions. What is the meaning of these vertex functions in terms of fermion operators?

The effective potential $U(\bar{\Delta}_0, \bar{\Delta}_5)$ is the generating functional of 1PI vertex functions at zero momentum. We can expand it in a Taylor series about $\bar{\Delta}_0 = \bar{\Delta}_5 = 0$:

$$U(\bar{\Delta}_0, \bar{\Delta}_5) = \sum_{n,m=0}^{\infty} \frac{1}{n!m!} \Gamma_{n,m}(0) \bar{\Delta}_0^n \bar{\Delta}_5^m,$$

where the coefficients $\Gamma_{n,m}(0)$ are the 1PI vertex functions with n external legs of $\hat{\Delta}_0$ and m external legs of $\hat{\Delta}_5$ at zero momentum. These vertex functions represent the multi-particle scattering amplitudes of the composite operators $\hat{\Delta}_0 = \bar{\psi}_a \psi_a$ and $\hat{\Delta}_5 = i\bar{\psi}_a \gamma_5 \psi_a$ mediated by the underlying fermion field.

(7) Find a perturbation theory expression for all coefficients of the effective potential U to leading order (i.e., the first nonvanishing term) in an expansion in terms of the coupling constant g. Do not do the integrals! Draw the Feynman diagram associated with each contribution.

To leading order in the coupling constant g (which corresponds to the one-loop level or the large-N limit), we introduce auxiliary fields $\sigma = -g\bar{\psi}\psi$ and $\pi = -ig\bar{\psi}\gamma_5\psi$. Integrating out the fermions in the path integral yields:

$$U(\bar{\Delta}_0, \bar{\Delta}_5) = \frac{\bar{\Delta}_0^2 + \bar{\Delta}_5^2}{2g} - N \int \frac{\mathrm{d}^2 k}{(2\pi)^2} \ln \det \left[\not{k} - (\bar{\Delta}_0 + i\gamma_5 \bar{\Delta}_5) \right].$$

Computing the determinant over the spinor space:

$$\det \left[\not{k} - (\bar{\Delta}_0 + i\gamma_5 \bar{\Delta}_5) \right] = k^2 + \bar{\Delta}_0^2 + \bar{\Delta}_5^2.$$

Thus, the effective potential is:

$$U(\bar{\Delta}_0, \bar{\Delta}_5) = \frac{\bar{\Delta}_0^2 + \bar{\Delta}_5^2}{2g} - N \int \frac{\mathrm{d}^2 k}{(2\pi)^2} \ln \left(k^2 + \bar{\Delta}_0^2 + \bar{\Delta}_5^2 \right).$$

The Taylor coefficients (vertex functions) $\Gamma_{n,m}$ are obtained by differentiating this potential with respect to $\bar{\Delta}_0$ and $\bar{\Delta}_5$. The corresponding Feynman diagrams are closed fermion loops with external insertions of the classical fields $\bar{\Delta}_0$ and $\bar{\Delta}_5$.

(8) Derive the chiral Ward identity for the generating functional of the vertex functions for the operators $\bar{\Delta}_0$ and $\bar{\Delta}_5$.

From the invariance of the path integral measure under the infinitesimal chiral transformation $\psi \to (1 + i\epsilon(x)\gamma_5)\psi$, we obtain the Schwinger-Dyson equation:

$$\partial_\mu J_5^\mu(x) = 2H_0(x)\hat{\Delta}_5(x) - 2H_5(x)\hat{\Delta}_0(x).$$

Integrating this over all space and taking the expectation value:

$$\int \mathrm{d}^2 x \left[2H_0(x)\langle\hat{\Delta}_5(x)\rangle - 2H_5(x)\langle\hat{\Delta}_0(x)\rangle \right] = 0.$$

By Legendre transform, the external fields are related to the derivatives of the effective potential:

$$H_0 = \frac{\partial U}{\partial \bar{\Delta}_0}, \quad H_5 = \frac{\partial U}{\partial \bar{\Delta}_5}.$$

Substituting these, we find the chiral Ward identity for the effective potential:

$$\bar{\Delta}_5 \frac{\partial U}{\partial \bar{\Delta}_0} - \bar{\Delta}_0 \frac{\partial U}{\partial \bar{\Delta}_5} = 0.$$

(9) Derive a Ward identity that relates the vertex functions Γ_{00} and Γ_{55} in the limit $p \to 0$. Assume that, as $H_0 \to 0$ and $H_g \to 0$, only $\bar{\Delta}_0$ has a nonzero expectation value (i.e., that the chiral symmetry is spontaneously broken).

We start with the Ward identity:

$$\bar{\Delta}_5 \frac{\partial U}{\partial \bar{\Delta}_0} - \bar{\Delta}_0 \frac{\partial U}{\partial \bar{\Delta}_5} = 0.$$

Differentiating this identity with respect to $\bar{\Delta}_5$ and setting $\bar{\Delta}_5 = 0$:

$$\frac{\partial U}{\partial \bar{\Delta}_0} - \bar{\Delta}_0 \frac{\partial^2 U}{\partial \bar{\Delta}_5^2} = 0.$$

Since $\frac{\partial U}{\partial \bar{\Delta}_0} = H_0$, and the second derivative is the 1PI vertex function $\Gamma_{55}(0) = \frac{\partial^2 U}{\partial \bar{\Delta}_5^2}$, we obtain:

$$\Gamma_{55}(0) = \frac{H_0}{\bar{\Delta}_0}.$$

In the limit where the explicit symmetry-breaking source $H_0 \to 0$, if the expectation value $\bar{\Delta}_0$ remains non-zero (spontaneous symmetry breaking), we have:

$$\Gamma_{55}(0) = 0.$$

This relates the pseudoscalar vertex function Γ_{55} to the scalar expectation value.

(10) Is there a Goldstone boson in this system? Justify your answer. Explain the significance of your answer for the original fermion problem. What does it say about its two-particle spectrum? And of the one-particle spectrum?

No, there is **no** physical Goldstone boson in this system. *Justification*: According to Coleman's theorem, continuous global symmetries cannot be spontaneously broken in 1+1 dimensions. This is because the infrared fluctuations of massless scalar fields in two dimensions are extremely strong and would lead to infinite fluctuations that restore the symmetry. Since the chiral symmetry cannot be spontaneously broken at the quantum level, the condition $\Gamma_{55}(0) = 0$ does not imply a physical massless Goldstone boson. *Significance for the original fermion problem*: 1.

> **One-particle spectrum**: The fermions acquire a dynamical mass through the mechanism of dimensional transmutation (large-N limit). Thus, the physical fermions are massive, but this mass generation does not break the global chiral symmetry. 2. **Two-particle spectrum**: Because the symmetry is not broken, there are no massless Goldstone bosons. The two-particle spectrum consists of massive bound states (mesons) of the fermion-antifermion pair, which are the physical states of the system.

References

[1]